Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Lima Ranch

Physical address of dairy:

Number and Street	City	County	Zip Code
13436 N Thornton RD	Lodi	San Joaquin	95242

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1932

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X055-X130-XX01-XXXX X055-X140-XX07-XXXX X055-X140-XX09-XXXX

B. OPERATORS

Hamm, Jack A

Operator name: Hamm, Jack A

Telephone no.: (209) 334-5422 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
13438 N Thornton RD	Lodi	CA	95242

This operator is responsible for paying permit fees.

C. OWNERS

Hamm, Jack A

Legal owner name: Hamm, Jack A

Telephone no.: (209) 334-5422 (Landline) (Cellular)

Mailing Address Number and Street	City	State	Zip Code
13438 N Thornton RD	Lodi	CA	95242

This owner is responsible for paying permit fees.

Hamm, Patty

Legal owner name: Hamm, Patty		Telephone no.:	(209) 334-5422	
			Landline	Cellular
13438 N Thornton RD	Lodi		CA	95242
Mailing Address Number and Street	City		State	Zip Code

AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	600	495	220	0
Number under roof	1,480	220	0	0	0	0
Maximum number	1,480	220	600	495	220	0
Average number	1,480	220	600	495	220	0
Avg live weight (lbs)	1,400	1,450	950	700		

Predominant milk cow breed: Holstein

Average milk production: 78 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 54,199.70 *tons per reporting period*

Total nitrogen from manure: 669,838.05 *lbs per reporting period*

After ammonia losses (30% loss applied): 468,886.64 *lbs per reporting period*

Total phosphorus from manure: 111,052.26 *lbs per reporting period*

Total potassium from manure: 291,212.54 *lbs per reporting period*

Total salt from manure: 747,447.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 23,996,042 *gallons*

Total nitrogen generated: 93,882.18 *lbs*

Total phosphorus generated: 15,610.98 *lbs*

Total potassium generated: 148,974.34 *lbs*

Total salt generated: 1,091,555.43 *lbs*

	23,996,042	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	23,996,042	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal Woodbridge	Ground water
IW 2	Ground water
IW 3	Ground water
IW 4	Ground water
IW 5	Ground water

Source Description	Type
IW 6	Ground water
IW 7	Ground water
IW Fontes	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
02/18/2023	Compost	2,100.00 *ton*	Dry-weight	17.9		25,300.00	7,300.00	6,400.00		45.30
08/21/2023	Separator solids	3,000.00 *ton*	Dry-weight	57.1		23,400.00	3,680.00	7,420.00		16.10
10/24/2023	Separator solids	2,000.00 *ton*	Dry-weight	57.1		23,400.00	3,680.00	7,420.00		16.10

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	187,625.46	40,959.06	53,900.28	2,252,724.60
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	187,625.46	40,959.06	53,900.28	2,252,724.60

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Allegre Ranch	15	15	0	none	X055-X160-XX23-XXXX X055-X160-XX25-XXXX X055-X160-XX43-XXXX
D1	42	42	2	process wastewater	X055-X140-XX04-XXXX X055-X140-XX05-XXXX
D2	54	54	2	process wastewater	X055-X140-XX07-XXXX X055-X140-XX08-XXXX X055-X140-XX09-XXXX
D3	34	34	2	process wastewater	X055-X130-XX01-XXXX X055-X130-XX02-XXXX
D4	34	34	1	process wastewater	X055-X130-XX02-XXXX
D5	13	13	1	none	X055-X140-XX10-XXXX
D6	17	17	1	none	X055-X130-XX01-XXXX
Fontes	60	60	1	none	X055-X180-XX03-XXXX
H1	23	23	2	manure	X055-X160-XX62-XXXX X055-X160-XX63-XXXX
H2	32	32	2	manure	X055-X160-XX62-XXXX
H3	30	30	2	manure	X055-X160-XX08-XXXX X055-X160-XX59-XXXX X055-X160-XX62-XXXX
Jennifer's	15	15	2	manure	X055-X160-XX21-XXXX X055-X160-XX45-XXXX
Mike's	17	17	2	manure	X055-X160-XX09-XXXX
RH1	62	62	2	manure	X055-X130-XX03-XXXX
RH2	56	56	2	both	X055-X130-XX03-XXXX
RH3	35	35	2	process wastewater	X055-X130-XX03-XXXX X055-X130-XX07-XXXX
RH4	19	19	2	process wastewater	X055-X130-XX07-XXXX
Totals for areas that were used for application	453	453	25		
Totals for areas that were not used for application	105	105	3		
Land application area totals	558	558	28		

B. CROPS AND HARVESTS

D1

Field name: D1

11/04/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 42 Plant date: 11/04/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/27/2023	607.00 *ton*	As-is		57.2	5,770.00	900.00	5,130.00		7.26

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.45	166.78	26.01	148.28	898.15

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 42 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	1,264.20 *ton*	As-is		67.6	3,400.00	640.00	3,300.00		4.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	30.10	204.68	38.53	198.66	858.21

D2

Field name: D2

10/17/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 54 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	810.00 *ton*	As-is		69.4	7,620.00	1,620.00	21,100.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	228.60	48.60	633.00	1,202.58

D2

06/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 54 Plant date: 06/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,744.20 *ton*	As-is		79.2	2,100.00	270.00	2,720.00		5.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	32.30	135.66	17.44	175.71	685.28

D3

Field name: D3

10/18/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 34 Plant date: 10/18/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/13/2023	506.60 *ton*	As-is		64.2	8,310.00	2,040.00	27,200.00		12.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.90	247.64	60.79	810.56	1,344.22

06/21/2023: Corn, silage

Crop: Corn, silage Acres planted: 34 Plant date: 06/21/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/13/2023	1,040.40 *ton*	As-is		78.4	2,530.00	270.00	2,250.00		3.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	30.60	154.84	16.52	137.70	515.55

D4

Field name: D4

10/17/2022: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 34 Plant date: 10/17/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/31/2023	278.80 *ton*	As-is		8.5	29,700.00	3,110.00	19,200.00		9.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	8.20	487.08	51.00	314.88	1,410.56

D5

Field name: D5

10/15/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 13 Plant date: 10/15/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	97.33 *ton*	As-is		8.2	28,900.00	2,810.00	43,300.00		13.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.49	432.74	42.08	648.37	1,800.72

D6

Field name: D6

D6

09/17/2020: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 17 Plant date: 09/17/2020

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/06/2023	163.40 *ton*	As-is		7.7	28,600.00	2,820.00	27,100.00		10.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	9.61	549.79	54.21	520.96	1,898.53

Fontes

Field name: Fontes

01/10/2018: Almond Field Run

Crop: Almond Field Run Acres planted: 60 Plant date: 01/10/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/28/2023	344.20 *ton*	As-is		18.7	15,800.00	1,230.00	16,700.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	5.00	215.00	36.50	232.50	0.00
Total actual harvest content	5.74	181.28	14.11	191.60	979.42

H1

Field name: H1

H1

10/22/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 23 Plant date: 10/22/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	326.60 *ton*	As-is		59.6	5,050.00	910.00	5,160.00		6.45

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.20	143.42	25.84	146.54	740.05

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 23 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	646.30 *ton*	As-is		64.6	3,790.00	640.00	3,930.00		4.00

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	28.10	213.00	35.97	220.87	795.79

H2

Field name: H2

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 32 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	451.20 *ton*	As-is		56.8	5,220.00	1,160.00	6,490.00		7.16

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.10	147.20	32.71	183.02	872.26

H2

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 32 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	937.60 *ton*	As-is		65.8	3,080.00	750.00	3,350.00		3.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	29.30	180.49	43.95	196.31	761.57

H3

Field name: H3

10/20/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 30 Plant date: 10/20/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	472.46 *ton*	As-is		61.6	5,200.00	1,100.00	5,720.00		6.55

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.75	163.79	34.65	180.16	792.22

06/07/2023: Corn, silage

Crop: Corn, silage Acres planted: 30 Plant date: 06/07/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/27/2023	961.14 *ton*	As-is		68.4	3,480.00	610.00	2,940.00		4.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	32.04	222.98	39.09	188.38	931.41

Jennifer's

Field name: Jennifer's

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 15 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	225.00 *ton*	As-is		63.6	4,960.00	1,020.00	5,480.00		7.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	148.80	30.60	164.40	773.14

06/03/2023: Corn, silage

Crop: Corn, silage Acres planted: 15 Plant date: 06/03/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	409.50 *ton*	As-is		67.4	3,390.00	520.00	2,350.00		3.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.30	185.09	28.39	128.31	605.19

Mike's

Field name: Mike's

10/21/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 17 Plant date: 10/21/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/26/2023	255.00 *ton*	As-is		63.6	4,960.00	1,020.00	5,480.00		7.08

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	15.00	148.80	30.60	164.40	773.14

Mike's

06/05/2023: Corn, silage

Crop: Corn, silage Acres planted: 17 Plant date: 06/05/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/26/2023	464.10 *ton*	As-is		67.4	3,390.00	520.00	2,350.00		3.40

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.30	185.09	28.39	128.31	605.19

RH1

Field name: RH1

10/13/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 62 Plant date: 10/13/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	1,176.33 *ton*	As-is		61.8	5,690.00	1,070.00	5,630.00		7.78

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	18.97	215.91	40.60	213.64	1,127.74

05/27/2023: Corn, silage

Crop: Corn, silage Acres planted: 62 Plant date: 05/27/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	2,008.80 *ton*	As-is		71.0	3,310.00	390.00	1,910.00		5.68

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	32.40	214.49	25.27	123.77	1,067.39

RH2

Field name: RH2

10/12/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 56 Plant date: 10/12/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/28/2023	822.00 *ton*	As-is		54.8	7,770.00	1,280.00	10,300.00		8.61

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	128.00	25.60	132.80	0.00
Total actual harvest content	14.68	228.11	37.58	302.38	1,142.50

05/23/2023: Corn, silage

Crop: Corn, silage Acres planted: 56 Plant date: 05/23/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	1,542.12 *ton*	As-is		69.2	3,910.00	490.00	4,730.00		6.59

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	27.54	215.35	26.99	260.51	1,117.88

RH3

Field name: RH3

10/11/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 35 Plant date: 10/11/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	520.80 *ton*	As-is		70.4	5,830.00	950.00	7,880.00		9.28

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.88	173.50	28.27	234.51	817.47

RH3

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	938.26 *ton*	As-is		67.0	4,220.00	430.00	4,800.00		6.89

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	26.81	226.25	23.05	257.35	1,219.04

RH4

Field name: RH4

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 19 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
04/29/2023	282.72 *ton*	As-is		68.2	5,340.00	990.00	8,060.00		9.52

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	14.88	158.92	29.46	239.87	900.94

05/22/2023: Corn, silage

Crop: Corn, silage Acres planted: 19 Plant date: 05/22/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/12/2023	365.28 *ton*	As-is		50.4	6,400.00	530.00	7,850.00		7.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	40.50	178.20	0.00
Total actual harvest content	19.23	246.08	20.38	301.84	1,449.43

NUTRIENT BUDGET

A. LAND APPLICATIONS

D1 - 11/04/2022: Oats, silage-soft dough

Field name: D1

Crop: Oats, silage-soft dough — Plant date: 11/04/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.85	13.07	176.15	970.34	1,013,208.00 *gal*
IW 2	Ground water	7.88	0.00	0.00	108.49	4,200,000.00 *gal*
Application event totals		137.73	13.07	176.15	1,078.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	80.54	2.27	23.79	625.93	1,050,084.00 *gal*
Application event totals		80.54	2.27	23.79	625.93	

D1 - 06/05/2023: Corn, silage

Field name: D1

Crop: Corn, silage — Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	152.23	18.05	165.17	1,673.93	1,017,492.00 *gal*
IW 2	Ground water	4.62	0.00	0.00	103.26	4,260,000.00 *gal*
Application event totals		156.85	18.05	165.17	1,777.20	

D1 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/05/2023	Banding	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Starter 4 - 0 - 0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	4.95	0.00	0.00	110.54	4,560,000.00 *gal*
Application event totals		4.95	0.00	0.00	110.54	

07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	4.88	0.00	0.00	109.08	4,500,000.00 *gal*
Application event totals		4.88	0.00	0.00	109.08	

07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	60.49	23.48	203.26	1,247.87	1,064,490.00 *gal*
IW 2	Ground water	3.65	0.00	0.00	81.45	3,360,000.00 *gal*
Application event totals		64.14	23.48	203.26	1,329.32	

08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	4.95	0.00	0.00	110.54	4,560,000.00 *gal*
Application event totals		4.95	0.00	0.00	110.54	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	4.82	0.00	0.00	107.63	4,440,000.00 *gal*
Application event totals		4.82	0.00	0.00	107.63	

D1 - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 2	Ground water	4.69	0.00	0.00	104.72	4,320,000.00 *gal*
Application event totals		4.69	0.00	0.00	104.72	

D2 - 10/17/2022: Oats, silage-soft dough

Field name: D2

Crop: Oats, silage-soft dough

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	86.25	2.44	25.47	670.31	1,445,850.00 *gal*
Application event totals		86.25	2.44	25.47	670.31	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	134.02	13.49	181.81	1,001.51	1,344,546.00 *gal*
IW 3	Ground water	7.46	0.00	0.00	136.53	5,700,000.00 *gal*
Application event totals		141.48	13.49	181.81	1,138.04	

D2 - 06/22/2023: Corn, silage

Field name: D2

Crop: Corn, silage

Plant date: 06/22/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

D2 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.95	0.00	0.00	163.84	6,840,000.00 *gal*
Application event totals		8.95	0.00	0.00	163.84	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.58	0.00	0.00	157.01	6,555,000.00 *gal*
Application event totals		8.58	0.00	0.00	157.01	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.21	26.86	232.56	1,427.78	1,565,946.00 *gal*
IW 3	Ground water	6.71	0.00	0.00	122.88	5,130,000.00 *gal*
Application event totals		75.93	26.86	232.56	1,550.66	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.21	0.00	0.00	150.19	6,270,000.00 *gal*
Application event totals		8.21	0.00	0.00	150.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.65	26.65	230.69	1,416.31	1,553,364.00 *gal*
IW 3	Ground water	6.71	0.00	0.00	122.88	5,130,000.00 *gal*
Application event totals		75.37	26.65	230.69	1,539.19	

D2 - 06/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.36	0.00	0.00	152.92	6,384,000.00 *gal*
Application event totals		8.36	0.00	0.00	152.92	

09/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 3	Ground water	8.21	0.00	0.00	150.19	6,270,000.00 *gal*
Application event totals		8.21	0.00	0.00	150.19	

D3 - 10/18/2022: Oats, silage-soft dough

Field name: D3

Crop: Oats, silage-soft dough Plant date: 10/18/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/01/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	147.75	14.87	200.44	1,104.12	933,300.00 *gal*
IW 6	Ground water	4.90	0.00	0.00	134.22	3,528,000.00 *gal*
Application event totals		152.65	14.87	200.44	1,238.34	

02/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	90.69	2.56	26.78	704.84	957,236.00 *gal*
Application event totals		90.69	2.56	26.78	704.84	

D3 - 06/21/2023: Corn, silage

Field name: D3

Crop: Corn, silage Plant date: 06/21/2023

D3 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.36	0.00	0.00	101.53	4,221,000.00 *gal*
Application event totals		3.36	0.00	0.00	101.53	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.46	0.00	0.00	104.56	4,347,000.00 *gal*
Application event totals		3.46	0.00	0.00	104.56	

06/21/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	69.16	26.84	232.39	1,426.75	985,252.00 *gal*
IW 6	Ground water	2.50	0.00	0.00	75.77	3,150,000.00 *gal*
Application event totals		71.67	26.84	232.39	1,502.51	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.36	0.00	0.00	101.53	4,221,000.00 *gal*
Application event totals		3.36	0.00	0.00	101.53	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	68.95	26.76	231.68	1,422.36	982,226.00 *gal*
IW 6	Ground water	2.56	0.00	0.00	77.28	3,213,000.00 *gal*
Application event totals		71.50	26.76	231.68	1,499.65	

D3 - 06/21/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.31	0.00	0.00	100.01	4,158,000.00 *gal*
Application event totals		3.31	0.00	0.00	100.01	

09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.21	0.00	0.00	96.98	4,032,000.00 *gal*
Application event totals		3.21	0.00	0.00	96.98	

D4 - 10/17/2022: Alfalfa, hay

Field name: D4

Crop: Alfalfa, hay

Plant date: 10/17/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/15/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	171.75	20.02	160.93	1,010.57	864,994.00 *gal*
IW 4	Ground water	7.15	0.00	0.00	141.37	3,600,000.00 *gal*
Application event totals		178.90	20.02	160.93	1,151.95	

04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	15.59	0.00	0.00	283.10	4,320,000.00 *gal*
Application event totals		15.59	0.00	0.00	283.10	

04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	12.77	0.00	0.00	231.99	3,540,000.00 *gal*
Application event totals		12.77	0.00	0.00	231.99	

D4 - 10/17/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	12.34	0.00	0.00	224.12	3,420,000.00 *gal*
Application event totals		12.34	0.00	0.00	224.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	13.21	0.00	0.00	239.85	3,660,000.00 *gal*
Application event totals		13.21	0.00	0.00	239.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	13.42	0.00	0.00	243.78	3,720,000.00 *gal*
Application event totals		13.42	0.00	0.00	243.78	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	57.54	22.33	193.35	1,187.07	819,740.00 *gal*
IW 4	Ground water	12.56	0.00	0.00	228.05	3,480,000.00 *gal*
Application event totals		70.10	22.33	193.35	1,415.12	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	12.99	0.00	0.00	235.92	3,600,000.00 *gal*
Application event totals		12.99	0.00	0.00	235.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	12.56	0.00	0.00	228.05	3,480,000.00 *gal*
Application event totals		12.56	0.00	0.00	228.05	

D4 - 10/17/2022: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 4	Ground water	12.34	0.00	0.00	224.12	3,420,000.00 *gal*
Application event totals		12.34	0.00	0.00	224.12	

D5 - 10/15/2019: Alfalfa, hay

Field name: D5

Crop: Alfalfa, hay

Plant date: 10/15/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.47	0.00	0.00	289.33	1,440,000.00 *gal*
Application event totals		5.47	0.00	0.00	289.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.36	0.00	0.00	283.30	1,410,000.00 *gal*
Application event totals		5.36	0.00	0.00	283.30	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.24	0.00	0.00	277.27	1,380,000.00 *gal*
Application event totals		5.24	0.00	0.00	277.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.47	0.00	0.00	289.33	1,440,000.00 *gal*
Application event totals		5.47	0.00	0.00	289.33	

D5 - 10/15/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.59	0.00	0.00	295.36	1,470,000.00 *gal*
Application event totals		5.59	0.00	0.00	295.36	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.70	0.00	0.00	301.38	1,500,000.00 *gal*
Application event totals		5.70	0.00	0.00	301.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.47	0.00	0.00	289.33	1,440,000.00 *gal*
Application event totals		5.47	0.00	0.00	289.33	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.24	0.00	0.00	277.27	1,380,000.00 *gal*
Application event totals		5.24	0.00	0.00	277.27	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.36	0.00	0.00	283.30	1,410,000.00 *gal*
Application event totals		5.36	0.00	0.00	283.30	

D6 - 09/17/2020: Alfalfa, hay

Field name: D6

Crop: Alfalfa, hay

Plant date: 09/17/2020

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

D6 - 09/17/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.58	0.00	0.00	295.00	1,920,000.00 *gal*
Application event totals		5.58	0.00	0.00	295.00	

04/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.23	0.00	0.00	276.56	1,800,000.00 *gal*
Application event totals		5.23	0.00	0.00	276.56	

05/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.41	0.00	0.00	285.78	1,860,000.00 *gal*
Application event totals		5.41	0.00	0.00	285.78	

07/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.41	0.00	0.00	285.78	1,860,000.00 *gal*
Application event totals		5.41	0.00	0.00	285.78	

07/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.58	0.00	0.00	295.00	1,920,000.00 *gal*
Application event totals		5.58	0.00	0.00	295.00	

08/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.49	0.00	0.00	290.39	1,890,000.00 *gal*
Application event totals		5.49	0.00	0.00	290.39	

D6 - 09/17/2020: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.49	0.00	0.00	290.39	1,890,000.00 *gal*
Application event totals		5.49	0.00	0.00	290.39	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.32	0.00	0.00	281.17	1,830,000.00 *gal*
Application event totals		5.32	0.00	0.00	281.17	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 5	Ground water	5.23	0.00	0.00	276.56	1,800,000.00 *gal*
Application event totals		5.23	0.00	0.00	276.56	

Fontes - 01/10/2018: Almond Field Run

Field name: Fontes

Crop: Almond Field Run

Plant date: 01/10/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	2.93	0.00	0.00	0.00	6,081,000.00 *gal*
Application event totals		2.93	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/30/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	2.93	0.00	0.00	0.00	6,081,000.00 *gal*
Application event totals		2.93	0.00	0.00	0.00	

Fontes - 01/10/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	40.80	0.00	0.00	0.00	
Application event totals		40.80	0.00	0.00	0.00	

04/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.40	0.00	0.00	578.97	6,033,000.00 *gal*
Application event totals		6.40	0.00	0.00	578.97	

04/20/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

05/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.40	0.00	0.00	578.97	6,033,000.00 *gal*
Application event totals		6.40	0.00	0.00	578.97	

05/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.40	0.00	0.00	578.40	6,027,000.00 *gal*
Application event totals		6.40	0.00	0.00	578.40	

05/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	38.40	0.00	0.00	0.00	
Application event totals		38.40	0.00	0.00	0.00	

Fontes - 01/10/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.40	0.00	0.00	578.40	6,027,000.00 *gal*
Application event totals		6.40	0.00	0.00	578.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.46	0.00	0.00	583.87	6,084,000.00 *gal*
Application event totals		6.46	0.00	0.00	583.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.46	0.00	0.00	583.87	6,084,000.00 *gal*
Application event totals		6.46	0.00	0.00	583.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.46	0.00	0.00	583.87	6,084,000.00 *gal*
Application event totals		6.46	0.00	0.00	583.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.46	0.00	0.00	583.87	6,084,000.00 *gal*
Application event totals		6.46	0.00	0.00	583.87	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.45	0.00	0.00	583.58	6,081,000.00 *gal*
Application event totals		6.45	0.00	0.00	583.58	

Fontes - 01/10/2018: Almond Field Run

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW Fontes	Ground water	6.45	0.00	0.00	583.58	6,081,000.00 *gal*
Application event totals		6.45	0.00	0.00	583.58	

11/01/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
CAN17	Solid commercial fertilizer	40.80	0.00	0.00	0.00	
Application event totals		40.80	0.00	0.00	0.00	

H1 - 10/22/2022: Oats, silage-soft dough

Field name: H1

Crop: Oats, silage-soft dough Plant date: 10/22/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Separator solids	191.41	43.68	72.54	1,305.48	340.00 *ton*
Application event totals		191.41	43.68	72.54	1,305.48	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	0.00	2,880,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

H1 - 06/03/2023: Corn, silage

Field name: H1

Crop: Corn, silage Plant date: 06/03/2023

H1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	205.70	89.25	119.85	4,896.00	250.00 *ton*
Application event totals		205.70	89.25	119.85	4,896.00	

05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	57.11	2,970,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.11	

06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.92	2,700,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.92	

06/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	50.19	2,610,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.19	

H1 - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	53.65	2,790,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	50.19	2,610,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	46.73	2,430,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	48.46	2,520,000.00 *gal*
Application event totals		0.00	0.00	0.00	48.46	

H2 - 10/21/2022: Oats, silage-soft dough

Field name: H2

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	194.22	44.32	73.60	1,324.68	480.00 *ton*
Application event totals		194.22	44.32	73.60	1,324.68	

H2 - 10/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	55.98	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.98	

H2 - 06/07/2023: Corn, silage

Field name: H2

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	201.07	87.24	117.15	4,785.84	340.00 *ton*
Application event totals		201.07	87.24	117.15	4,785.84	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	57.22	4,140,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.22	

06/07/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	52.24	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	52.24	

H2 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.00	3,690,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.00	

07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	55.98	4,050,000.00 *gal*
Application event totals		0.00	0.00	0.00	55.98	

08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	53.49	3,870,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.49	

08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	49.76	3,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.76	

09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	47.27	3,420,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.27	

H3 - 10/20/2022: Oats, silage-soft dough

H3 - 10/20/2022: Oats, silage-soft dough

Field name: H3

Crop: Oats, silage-soft dough

Plant date: 10/20/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	194.22	44.32	73.60	1,324.68	450.00 *ton*
Application event totals		194.22	44.32	73.60	1,324.68	

09/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	0.00	3,780,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

H3 - 06/07/2023: Corn, silage

Field name: H3

Crop: Corn, silage

Plant date: 06/07/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	208.17	90.32	121.29	4,954.75	330.00 *ton*
Application event totals		208.17	90.32	121.29	4,954.75	

05/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	57.05	3,870,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.05	

H3 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	54.40	3,690,000.00 *gal*
Application event totals		0.00	0.00	0.00	54.40	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.75	3,510,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	53.07	3,600,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	50.42	3,420,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.42	

H3 - 06/07/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	49.09	3,330,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.09	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	46.44	3,150,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.44	

Jennifer's - 10/21/2022: Oats, silage-soft dough

Field name: Jennifer's

Crop: Oats, silage-soft dough Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	189.90	43.34	71.97	1,295.24	220.00 *ton*
Application event totals		189.90	43.34	71.97	1,295.24	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	0.00	1,890,000.00 *gal*
Application event totals		0.00	0.00	0.00	0.00	

Jennifer's - 06/03/2023: Corn, silage

Field name: Jennifer's

Crop: Corn, silage Plant date: 06/03/2023

Jennifer's - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	201.86	87.58	117.61	4,804.61	160.00 *ton*
Application event totals		201.86	87.58	117.61	4,804.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	57.05	1,935,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.05	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/03/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	50.42	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	50.42	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	49.09	1,665,000.00 *gal*
Application event totals		0.00	0.00	0.00	49.09	

Jennifer's - 06/03/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	53.07	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.07	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.75	1,755,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.75	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	47.77	1,620,000.00 *gal*
Application event totals		0.00	0.00	0.00	47.77	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	45.11	1,530,000.00 *gal*
Application event totals		0.00	0.00	0.00	45.11	

Mike's - 10/21/2022: Oats, silage-soft dough

Field name: Mike's

Crop: Oats, silage-soft dough

Plant date: 10/21/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	190.41	43.45	72.16	1,298.71	250.00 *ton*
Application event totals		190.41	43.45	72.16	1,298.71	

Mike's - 10/21/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	56.20	2,160,000.00 *gal*
Application event totals		0.00	0.00	0.00	56.20	

Mike's - 06/05/2023: Corn, silage

Field name: Mike's

Crop: Corn, silage

Plant date: 06/05/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	200.38	86.94	116.75	4,769.28	180.00 *ton*
Application event totals		200.38	86.94	116.75	4,769.28	

05/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	57.37	2,205,000.00 *gal*
Application event totals		0.00	0.00	0.00	57.37	

06/05/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.51	1,980,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.51	

Mike's - 06/05/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/25/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	46.83	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	53.85	2,070,000.00 *gal*
Application event totals		0.00	0.00	0.00	53.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	51.51	1,980,000.00 *gal*
Application event totals		0.00	0.00	0.00	51.51	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	46.83	1,800,000.00 *gal*
Application event totals		0.00	0.00	0.00	46.83	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal Woodbridge	Ground water	0.00	0.00	0.00	44.49	1,710,000.00 *gal*
Application event totals		0.00	0.00	0.00	44.49	

RH1 - 10/13/2022: Oats, silage-soft dough

RH1 - 10/13/2022: Oats, silage-soft dough

Field name: RH1

Crop: Oats, silage-soft dough

Plant date: 10/13/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	229.72	52.42	87.06	1,566.83	1,100.00 *ton*
Application event totals		229.72	52.42	87.06	1,566.83	

09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.78	0.00	0.00	126.47	7,830,000.00 *gal*
Application event totals		5.78	0.00	0.00	126.47	

RH1 - 05/27/2023: Corn, silage

Field name: RH1

Crop: Corn, silage

Plant date: 05/27/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/06/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.77	0.00	0.00	100.43	7,938,000.00 *gal*
Application event totals		3.77	0.00	0.00	100.43	

05/01/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	213.66	92.70	124.49	5,085.52	700.00 *ton*
Application event totals		213.66	92.70	124.49	5,085.52	

RH1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/27/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.75	0.00	0.00	99.75	7,884,000.00 *gal*
Application event totals		3.75	0.00	0.00	99.75	

06/16/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN 32	Solid commercial fertilizer	32.00	0.00	0.00	0.00	
Application event totals		32.00	0.00	0.00	0.00	

07/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.73	0.00	0.00	99.41	7,857,000.00 *gal*
Application event totals		3.73	0.00	0.00	99.41	

07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.72	0.00	0.00	99.07	7,830,000.00 *gal*
Application event totals		3.72	0.00	0.00	99.07	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.64	0.00	0.00	97.02	7,668,000.00 *gal*
Application event totals		3.64	0.00	0.00	97.02	

RH1 - 05/27/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.62	0.00	0.00	96.33	7,614,000.00 *gal*
Application event totals		3.62	0.00	0.00	96.33	

RH2 - 10/12/2022: Oats, silage-soft dough

Field name: RH2

Crop: Oats, silage-soft dough Plant date: 10/12/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Separator Solids	Separator solids	231.21	52.76	87.62	1,577.00	1,000.00 *ton*
Application event totals		231.21	52.76	87.62	1,577.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	5.73	0.00	0.00	125.53	7,020,000.00 *gal*
Application event totals		5.73	0.00	0.00	125.53	

RH2 - 05/23/2023: Corn, silage

Field name: RH2

Crop: Corn, silage Plant date: 05/23/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

RH2 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.81	0.00	0.00	101.36	7,236,000.00 *gal*
Application event totals		3.81	0.00	0.00	101.36	

05/23/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.69	0.00	0.00	98.33	7,020,000.00 *gal*
Application event totals		3.69	0.00	0.00	98.33	

06/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	161.26	19.12	174.97	1,773.23	1,437,128.00 *gal*
IW 7	Ground water	2.56	0.00	0.00	68.08	4,860,000.00 *gal*
Application event totals		163.82	19.12	174.97	1,841.30	

07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	2.56	0.00	0.00	68.08	4,860,000.00 *gal*
Application event totals		2.56	0.00	0.00	68.08	

07/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	61.34	23.81	206.10	1,265.35	1,439,200.00 *gal*
IW 7	Ground water	2.35	0.00	0.00	70.97	4,860,000.00 *gal*
Application event totals		63.68	23.81	206.10	1,336.33	

RH2 - 05/23/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.61	0.00	0.00	96.06	6,858,000.00 *gal*
Application event totals		3.61	0.00	0.00	96.06	

08/22/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 7	Ground water	3.55	0.00	0.00	94.55	6,750,000.00 *gal*
Application event totals		3.55	0.00	0.00	94.55	

RH3 - 10/11/2022: Oats, silage-soft dough

Field name: RH3

Crop: Oats, silage-soft dough Plant date: 10/11/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	132.17	13.30	179.30	987.67	859,425.00 *gal*
IW 6	Ground water	5.10	0.00	0.00	139.70	3,780,000.00 *gal*
Application event totals		137.27	13.30	179.30	1,127.37	

02/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	83.35	2.35	24.62	647.83	905,695.00 *gal*
Application event totals		83.35	2.35	24.62	647.83	

RH3 - 05/22/2023: Corn, silage

Field name: RH3

Crop: Corn, silage Plant date: 05/22/2023

RH3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	162.41	19.26	176.21	1,785.87	904,610.00 *gal*
IW 6	Ground water	2.92	0.00	0.00	88.32	3,780,000.00 *gal*
Application event totals		165.33	19.26	176.21	1,874.19	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.41	0.00	0.00	103.04	4,410,000.00 *gal*
Application event totals		3.41	0.00	0.00	103.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.31	0.00	0.00	100.10	4,284,000.00 *gal*
Application event totals		3.31	0.00	0.00	100.10	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.64	24.70	213.84	1,312.86	933,275.00 *gal*
IW 6	Ground water	2.43	0.00	0.00	73.60	3,150,000.00 *gal*
Application event totals		66.07	24.70	213.84	1,386.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.26	0.00	0.00	98.63	4,221,000.00 *gal*
Application event totals		3.26	0.00	0.00	98.63	

RH3 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.21	0.00	0.00	97.16	4,158,000.00 *gal*
Application event totals		3.21	0.00	0.00	97.16	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.11	0.00	0.00	94.21	4,032,000.00 *gal*
Application event totals		3.11	0.00	0.00	94.21	

RH4 - 10/10/2022: Oats, silage-soft dough

Field name: RH4

Crop: Oats, silage-soft dough Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	78.03	2.20	23.04	606.42	460,237.00 *gal*
Application event totals		78.03	2.20	23.04	606.42	

09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	129.77	13.06	176.05	969.77	458,090.00 *gal*
IW 6	Ground water	5.01	0.00	0.00	137.24	2,016,000.00 *gal*
Application event totals		134.78	13.06	176.05	1,107.02	

RH4 - 05/22/2023: Corn, silage

Field name: RH4

Crop: Corn, silage Plant date: 05/22/2023

RH4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/03/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	162.67	19.29	176.50	1,788.77	491,872.00 *gal*
IW 6	Ground water	2.96	0.00	0.00	89.49	2,079,000.00 *gal*
Application event totals		165.63	19.29	176.50	1,878.26	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/22/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.77	0.00	0.00	113.89	2,646,000.00 *gal*
Application event totals		3.77	0.00	0.00	113.89	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.59	0.00	0.00	108.47	2,520,000.00 *gal*
Application event totals		3.59	0.00	0.00	108.47	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/10/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.41	0.00	0.00	103.04	2,394,000.00 *gal*
Application event totals		3.41	0.00	0.00	103.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	63.91	24.80	214.75	1,318.43	508,782.00 *gal*
IW 6	Ground water	2.69	0.00	0.00	81.35	1,890,000.00 *gal*
Application event totals		66.60	24.80	214.75	1,399.78	

RH4 - 05/22/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/07/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.41	0.00	0.00	103.04	2,394,000.00 *gal*
Application event totals		3.41	0.00	0.00	103.04	

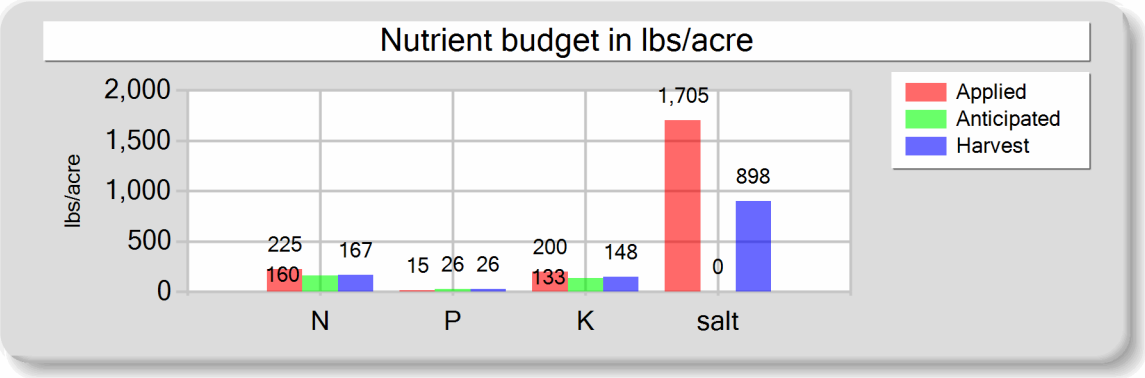
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
IW 6	Ground water	3.32	0.00	0.00	100.33	2,331,000.00 *gal*
Application event totals		3.32	0.00	0.00	100.33	

B. NUTRIENT BUDGET

D1 - 11/04/2022: Oats, silage-soft dough

Field name: D1 Crop: Oats, silage-soft dough Plant date: 11/04/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	210.38	15.34	199.94	1,596.26
Fresh water	7.88	0.00	0.00	108.49
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	225.26	15.34	199.94	1,704.75
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	166.78	26.01	148.28	898.15
Nutrient balance	58.48	-10.67	51.65	806.60
Applied to removed ratio	1.35	0.59	1.35	1.90

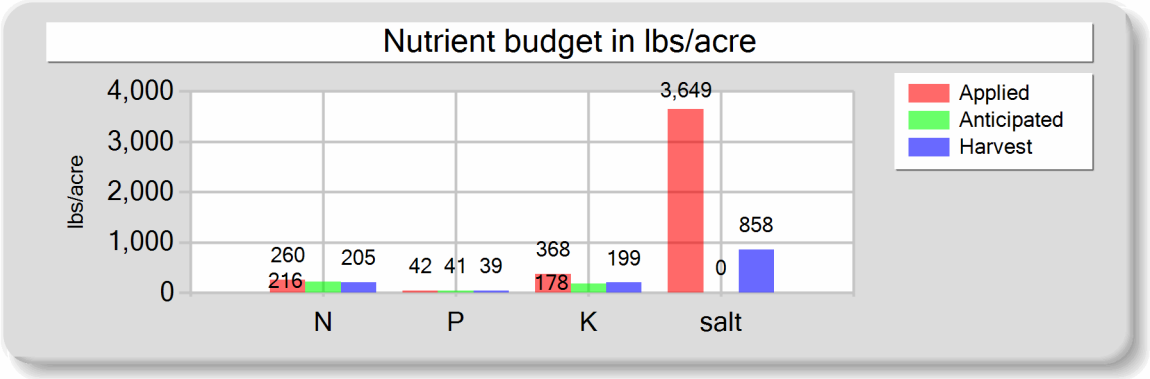
Fresh water applied	
4,200,000.00	*gallons*
154.67	*acre-inches*
3.68	*inches/acre*

Process wastewater applied	
2,063,292.00	*gallons*
75.98	*acre-inches*
1.81	*inches/acre*

Total harvests for the crop	
1	*harvests*

D1 - 06/05/2023: Corn, silage

Field name: D1 Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	212.72	41.53	368.42	2,921.81
Fresh water	32.55	0.00	0.00	727.21
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	260.27	41.53	368.42	3,649.02
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	204.68	38.53	198.66	858.21
Nutrient balance	55.59	3.00	169.76	2,790.80
Applied to removed ratio	1.27	1.08	1.85	4.25

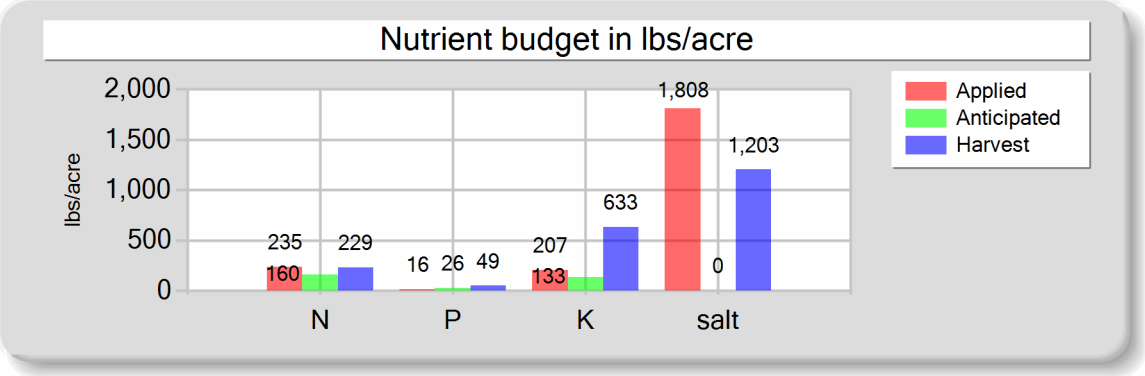
Fresh water applied	
30,000,000.00	*gallons*
1,104.80	*acre-inches*
26.30	*inches/acre*

Process wastewater applied	
2,081,982.00	*gallons*
76.67	*acre-inches*
1.83	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2 - 10/17/2022: Oats, silage-soft dough

Field name: D2 Crop: Oats, silage-soft dough Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	220.27	15.92	207.28	1,671.82
Fresh water	7.46	0.00	0.00	136.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	234.73	15.92	207.28	1,808.36
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	228.60	48.60	633.00	1,202.58
Nutrient balance	6.13	-32.68	-425.72	605.78
Applied to removed ratio	1.03	0.33	0.33	1.50

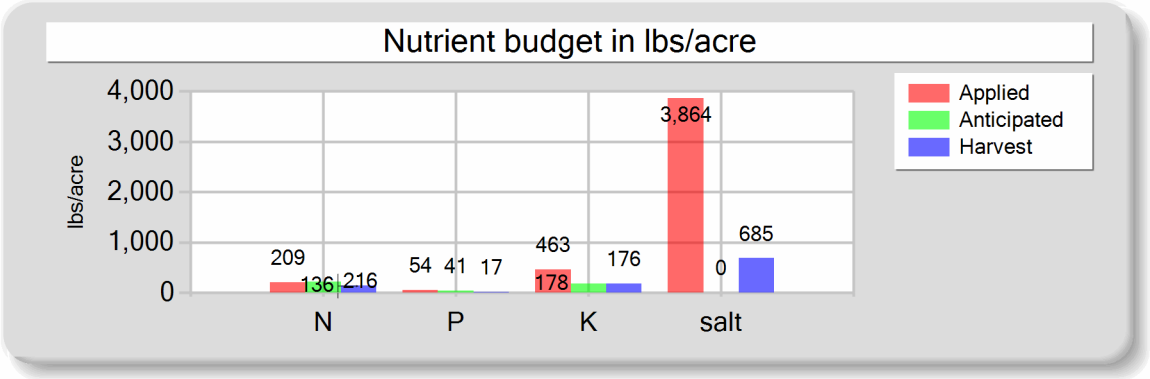
Fresh water applied	
5,700,000.00	*gallons*
209.91	*acre-inches*
3.89	*inches/acre*

Process wastewater applied	
2,790,396.00	*gallons*
102.76	*acre-inches*
1.90	*inches/acre*

Total harvests for the crop	
1	*harvests*

D2 - 06/22/2023: Corn, silage

Field name: D2 Crop: Corn, silage Plant date: 06/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	137.87	53.51	463.25	2,844.09
Fresh water	55.73	0.00	0.00	1,019.91
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	208.60	53.51	463.25	3,863.99
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	135.66	17.44	175.71	685.28
Nutrient balance	72.94	36.07	287.54	3,178.72
Applied to removed ratio	1.54	3.07	2.64	5.64

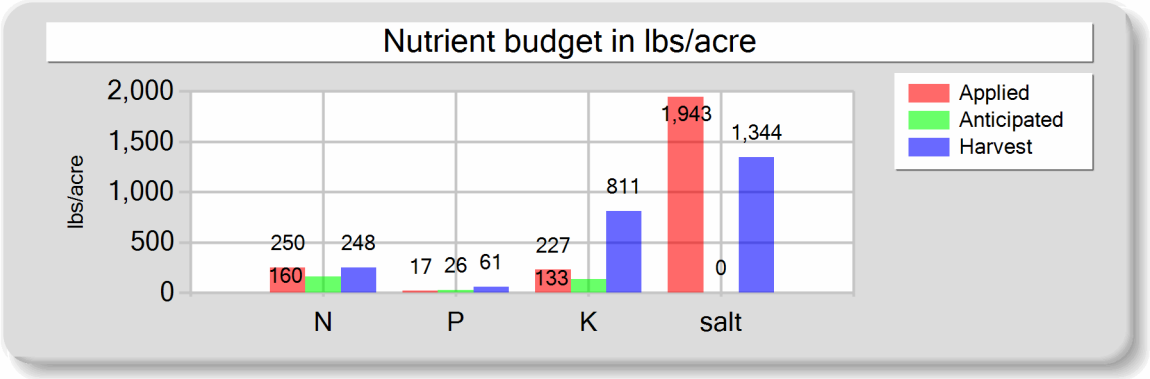
Fresh water applied	
42,579,000.00	*gallons*
1,568.04	*acre-inches*
29.04	*inches/acre*

Process wastewater applied	
3,119,310.00	*gallons*
114.87	*acre-inches*
2.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

D3 - 10/18/2022: Oats, silage-soft dough

Field name: D3 Crop: Oats, silage-soft dough Plant date: 10/18/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	238.44	17.43	227.22	1,808.95
Fresh water	4.90	0.00	0.00	134.22
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	250.34	17.43	227.22	1,943.17
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	247.64	60.79	810.56	1,344.22
Nutrient balance	2.70	-43.36	-583.34	598.95
Applied to removed ratio	1.01	0.29	0.28	1.45

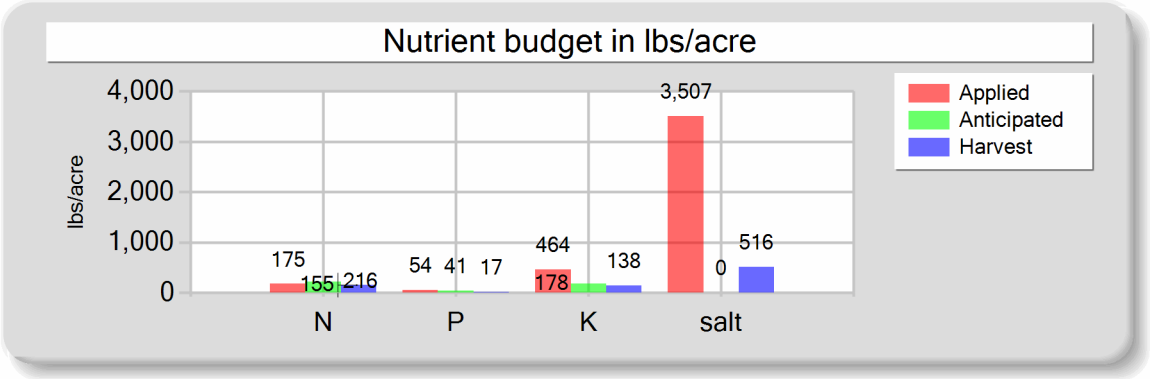
Fresh water applied	
3,528,000.00	*gallons*
129.92	*acre-inches*
3.82	*inches/acre*

Process wastewater applied	
1,890,536.00	*gallons*
69.62	*acre-inches*
2.05	*inches/acre*

Total harvests for the crop	
1	*harvests*

D3 - 06/21/2023: Corn, silage

Field name: D3 Crop: Corn, silage Plant date: 06/21/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	138.11	53.60	464.07	2,849.11
Fresh water	21.74	0.00	0.00	657.66
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.85	53.60	464.07	3,506.77
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	154.84	16.52	137.70	515.55
Nutrient balance	20.02	37.08	326.37	2,991.23
Applied to removed ratio	1.13	3.24	3.37	6.80

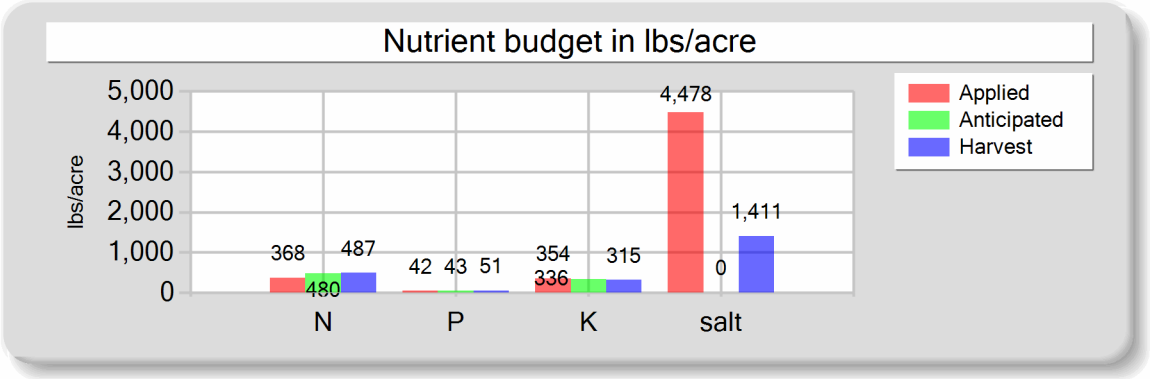
Fresh water applied	
27,342,000.00	*gallons*
1,006.91	*acre-inches*
29.62	*inches/acre*

Process wastewater applied	
1,967,478.00	*gallons*
72.46	*acre-inches*
2.13	*inches/acre*

Total harvests for the crop	
1	*harvests*

D4 - 10/17/2022: Alfalfa, hay

Field name: D4 Crop: Alfalfa, hay Plant date: 10/17/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	229.30	42.35	354.28	2,197.64
Fresh water	124.91	0.00	0.00	2,280.36
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	368.21	42.35	354.28	4,478.00
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	487.08	51.00	314.88	1,410.56
Nutrient balance	-118.87	-8.65	39.40	3,067.44
Applied to removed ratio	0.76	0.83	1.13	3.17

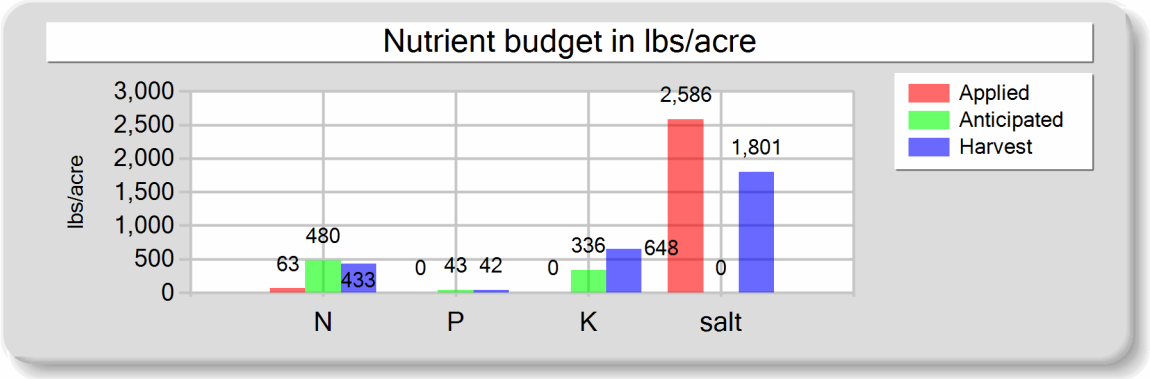
Fresh water applied	
36,240,000.00	*gallons*
1,334.60	*acre-inches*
39.25	*inches/acre*

Process wastewater applied	
1,684,734.00	*gallons*
62.04	*acre-inches*
1.82	*inches/acre*

Total harvests for the crop	
1	*harvests*

D5 - 10/15/2019: Alfalfa, hay

Field name: D5 Crop: Alfalfa, hay Plant date: 10/15/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	48.91	0.00	0.00	2,585.87
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	62.91	0.00	0.00	2,585.87
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	432.74	42.08	648.37	1,800.72
Nutrient balance	-369.84	-42.08	-648.37	785.14
Applied to removed ratio	0.15	0.00	0.00	1.44

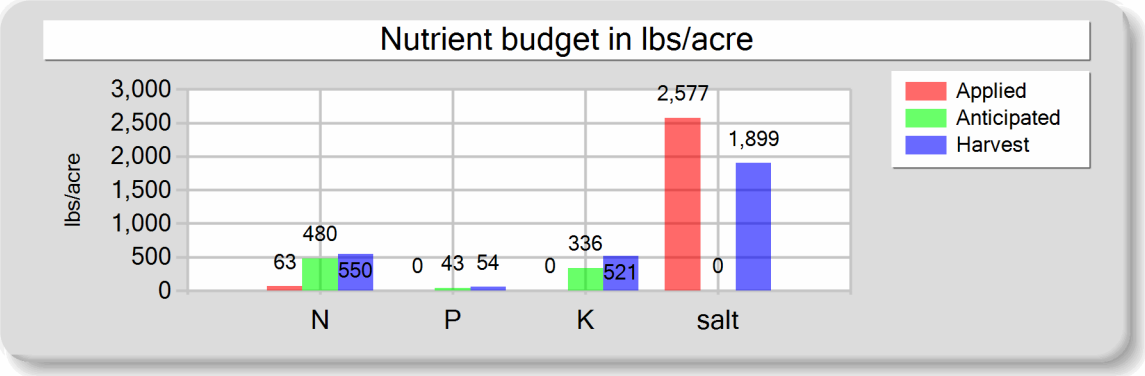
Fresh water applied	
12,870,000.00	*gallons*
473.96	*acre-inches*
36.46	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

D6 - 09/17/2020: Alfalfa, hay

Field name: D6 Crop: Alfalfa, hay Plant date: 09/17/2020



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	48.73	0.00	0.00	2,576.65
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	62.73	0.00	0.00	2,576.65
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	549.79	54.21	520.96	1,898.53
Nutrient balance	-487.06	-54.21	-520.96	678.11
Applied to removed ratio	0.11	0.00	0.00	1.36

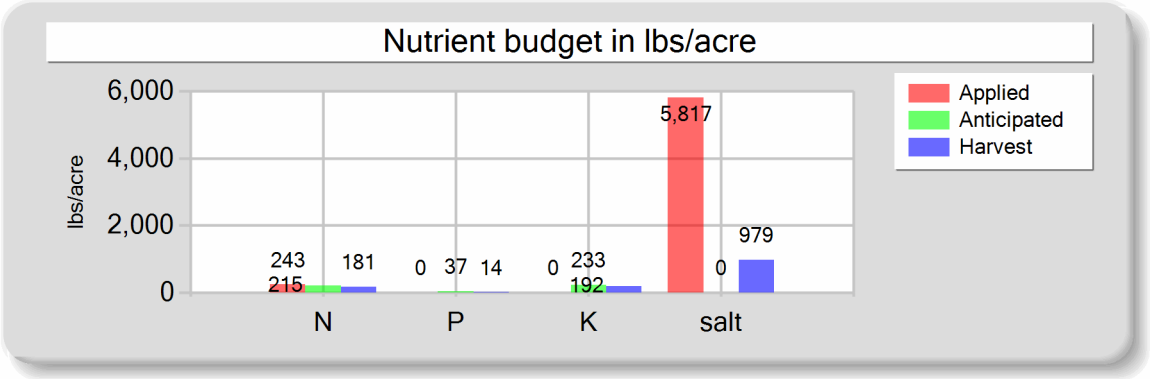
Fresh water applied	
16,770,000.00	*gallons*
617.58	*acre-inches*
36.33	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Fontes - 01/10/2018: Almond Field Run

Field name: Fontes Crop: Almond Field Run Plant date: 01/10/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	158.40	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	70.20	0.00	0.00	5,817.36
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	242.60	0.00	0.00	5,817.36
Anticipated crop nutrient removal	215.00	36.50	232.50	0.00
Actual crop nutrient removal	181.28	14.11	191.60	979.42
Nutrient balance	61.32	-14.11	-191.60	4,837.94
Applied to removed ratio	1.34	0.00	0.00	5.94

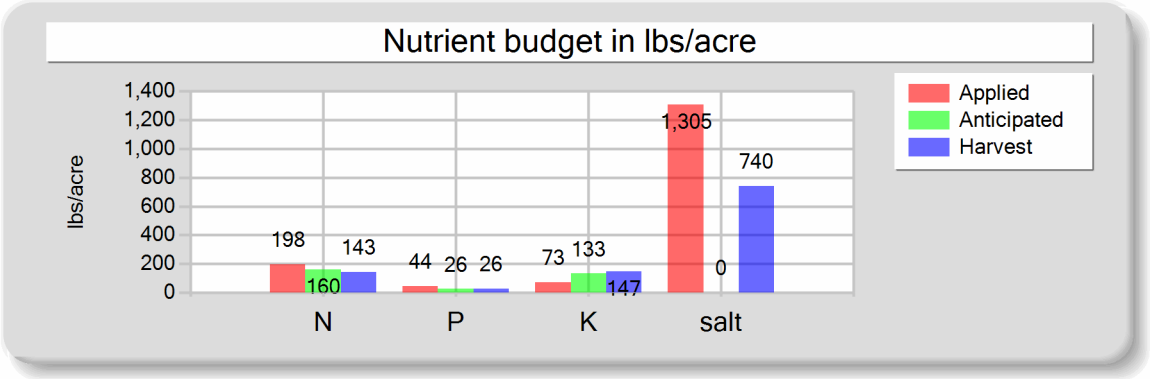
Fresh water applied	
72,780,000.00	*gallons*
2,680.24	*acre-inches*
44.67	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 10/22/2022: Oats, silage-soft dough

Field name: H1 Crop: Oats, silage-soft dough Plant date: 10/22/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	191.41	43.68	72.54	1,305.48
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	198.41	43.68	72.54	1,305.48
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	143.42	25.84	146.54	740.05
Nutrient balance	54.99	17.84	-74.01	565.43
Applied to removed ratio	1.38	1.69	0.49	1.76

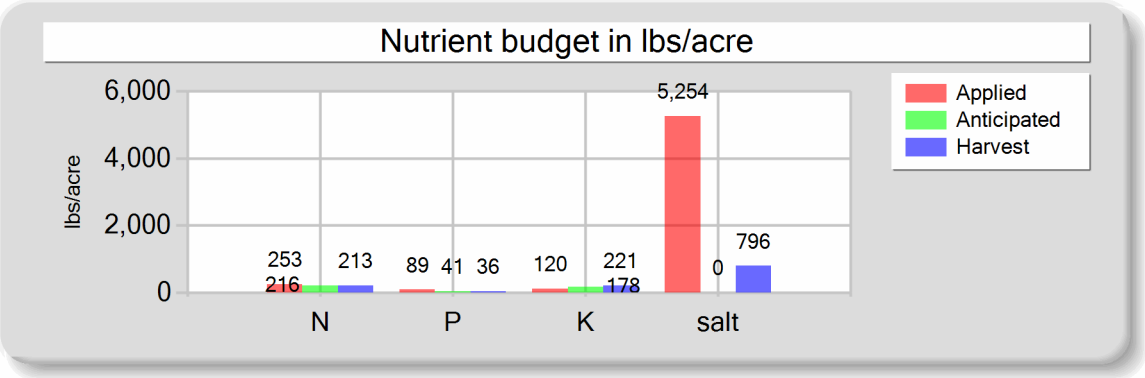
Fresh water applied	
2,880,000.00	*gallons*
106.06	*acre-inches*
4.61	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H1 - 06/03/2023: Corn, silage

Field name: H1 Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	205.70	89.25	119.85	4,896.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	358.25
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	252.70	89.25	119.85	5,254.25
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	213.00	35.97	220.87	795.79
Nutrient balance	39.70	53.28	-101.02	4,458.46
Applied to removed ratio	1.19	2.48	0.54	6.60

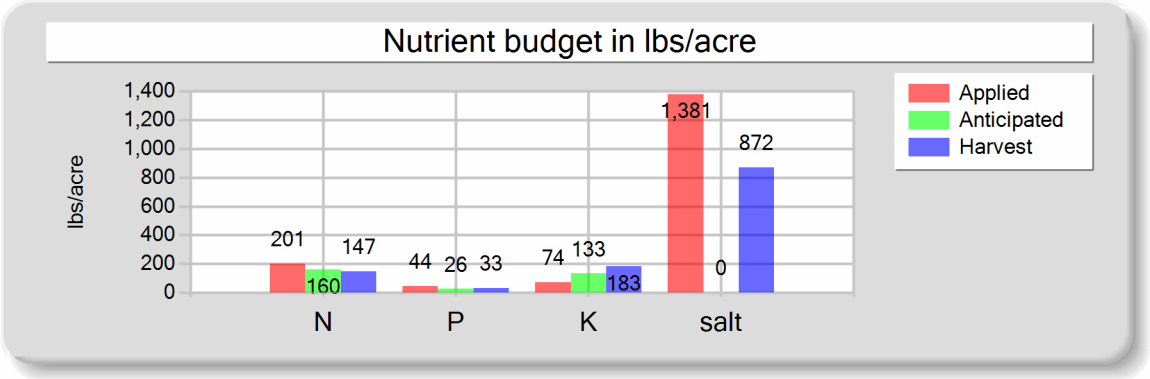
Fresh water applied	
18,630,000.00	*gallons*
686.08	*acre-inches*
29.83	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H2 - 10/21/2022: Oats, silage-soft dough

Field name: H2 Crop: Oats, silage-soft dough Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	194.22	44.32	73.60	1,324.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	55.98
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.22	44.32	73.60	1,380.66
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	147.20	32.71	183.02	872.26
Nutrient balance	54.02	11.61	-109.41	508.40
Applied to removed ratio	1.37	1.35	0.40	1.58

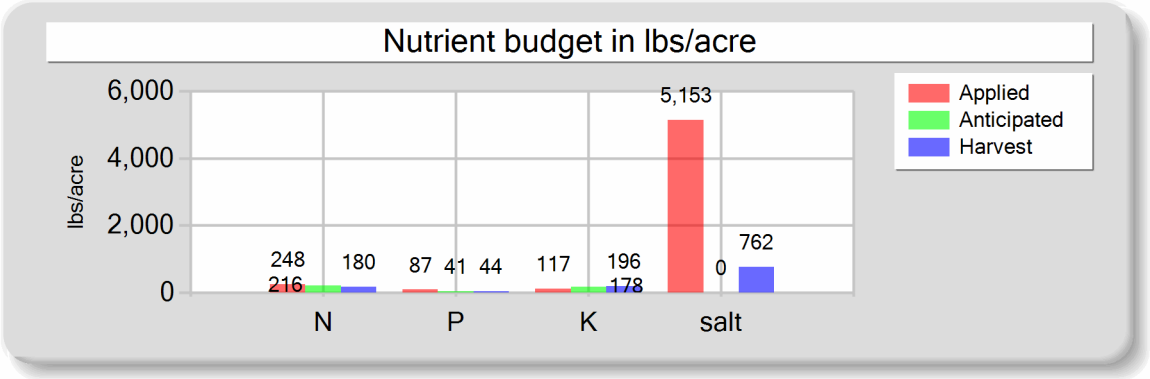
Fresh water applied	
4,050,000.00	*gallons*
149.15	*acre-inches*
4.66	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H2 - 06/07/2023: Corn, silage

Field name: H2 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	201.07	87.24	117.15	4,785.84
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	366.96
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.07	87.24	117.15	5,152.80
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	180.49	43.95	196.31	761.57
Nutrient balance	67.58	43.29	-79.16	4,391.23
Applied to removed ratio	1.37	1.99	0.60	6.77

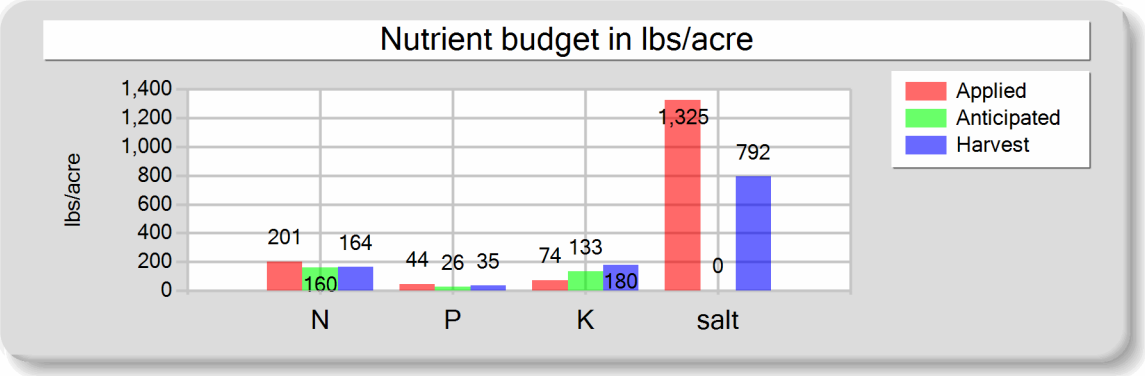
Fresh water applied	
26,550,000.00	*gallons*
977.75	*acre-inches*
30.55	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H3 - 10/20/2022: Oats, silage-soft dough

Field name: H3 Crop: Oats, silage-soft dough Plant date: 10/20/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	194.22	44.32	73.60	1,324.68
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	201.22	44.32	73.60	1,324.68
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	163.79	34.65	180.16	792.22
Nutrient balance	37.43	9.67	-106.56	532.46
Applied to removed ratio	1.23	1.28	0.41	1.67

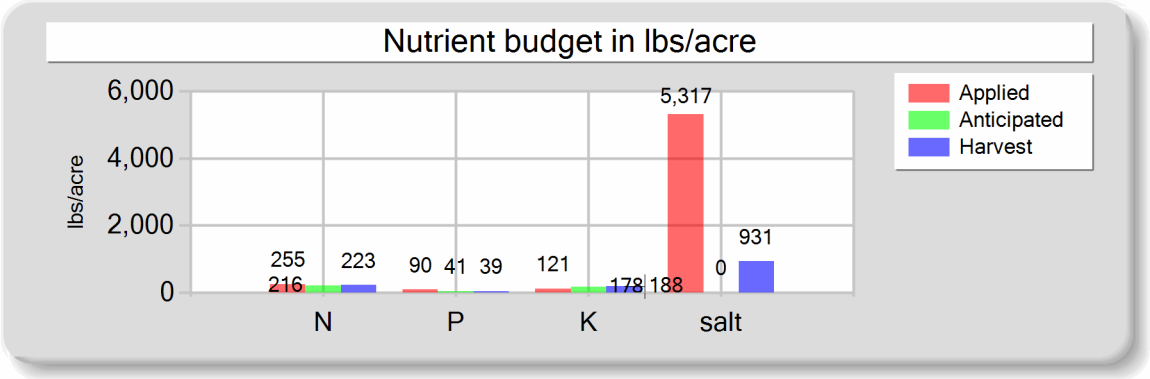
Fresh water applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
4.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

H3 - 06/07/2023: Corn, silage

Field name: H3 Crop: Corn, silage Plant date: 06/07/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	208.17	90.32	121.29	4,954.75
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	362.23
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	255.17	90.32	121.29	5,316.98
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	222.98	39.09	188.38	931.41
Nutrient balance	32.18	51.23	-67.10	4,385.57
Applied to removed ratio	1.14	2.31	0.64	5.71

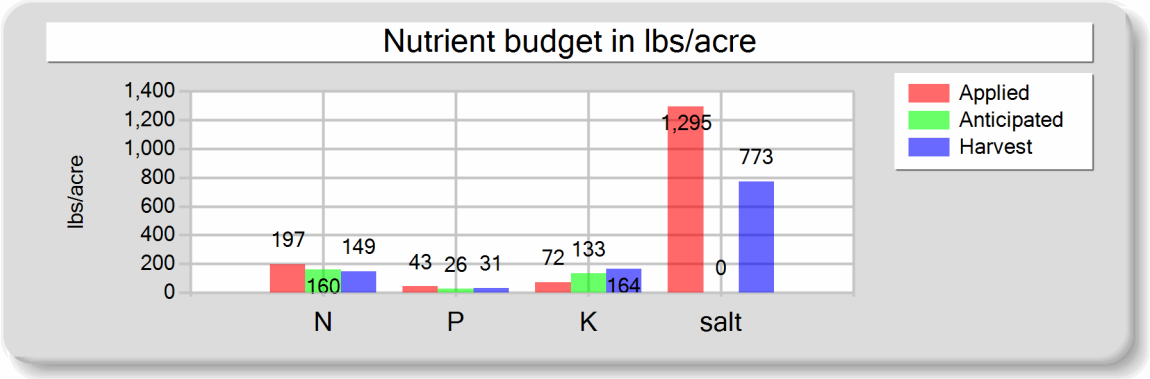
Fresh water applied	
24,570,000.00	*gallons*
904.83	*acre-inches*
30.16	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jennifer's - 10/21/2022: Oats, silage-soft dough

Field name: Jennifer's | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	189.90	43.34	71.97	1,295.24
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	196.90	43.34	71.97	1,295.24
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.80	30.60	164.40	773.14
Nutrient balance	48.10	12.74	-92.43	522.11
Applied to removed ratio	1.32	1.42	0.44	1.68

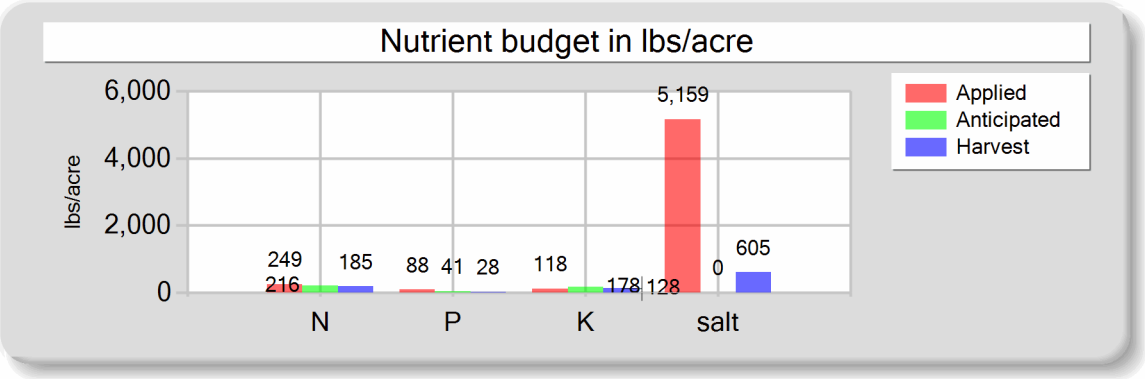
Fresh water applied	
1,890,000.00	*gallons*
69.60	*acre-inches*
4.64	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Jennifer's - 06/03/2023: Corn, silage

Field name: Jennifer's Crop: Corn, silage Plant date: 06/03/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	201.86	87.58	117.61	4,804.61
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	354.27
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	248.86	87.58	117.61	5,158.88
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	185.09	28.39	128.31	605.19
Nutrient balance	63.77	59.19	-10.70	4,553.69
Applied to removed ratio	1.34	3.08	0.92	8.52

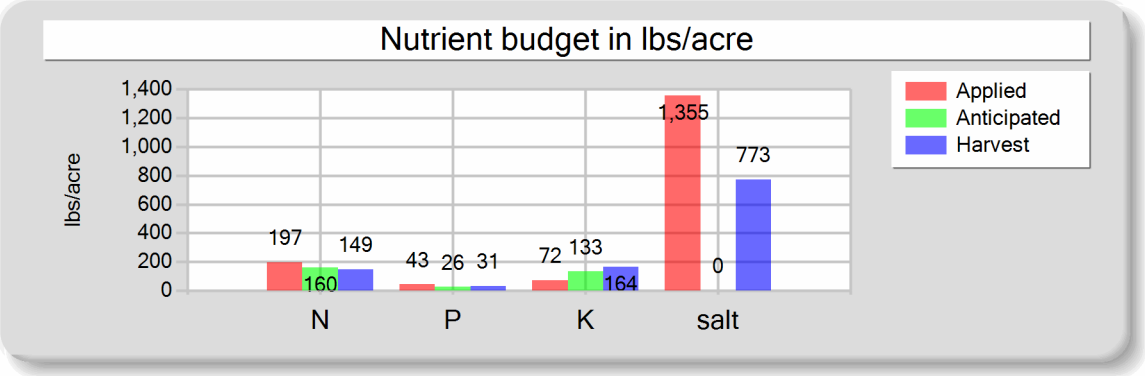
Fresh water applied	
12,015,000.00	*gallons*
442.47	*acre-inches*
29.50	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mike's - 10/21/2022: Oats, silage-soft dough

Field name: Mike's | Crop: Oats, silage-soft dough | Plant date: 10/21/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	190.41	43.45	72.16	1,298.71
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	56.20
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	197.41	43.45	72.16	1,354.90
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	148.80	30.60	164.40	773.14
Nutrient balance	48.61	12.85	-92.24	581.77
Applied to removed ratio	1.33	1.42	0.44	1.75

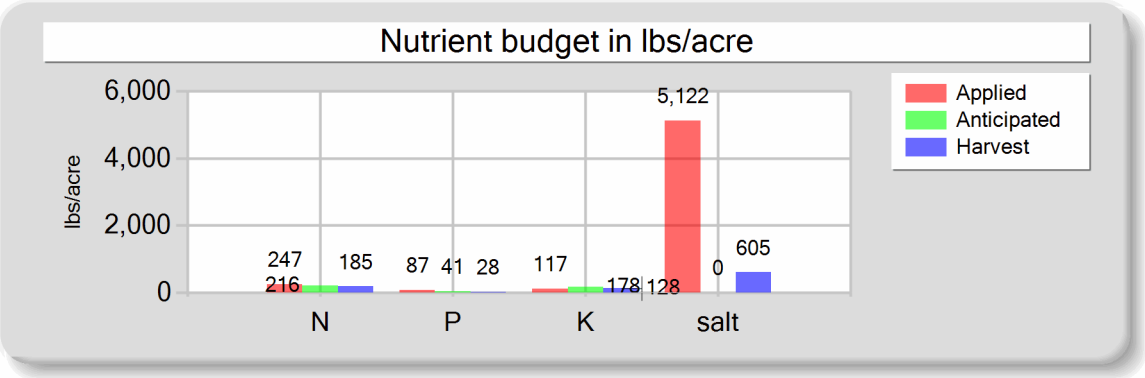
Fresh water applied	
2,160,000.00	*gallons*
79.55	*acre-inches*
4.68	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

Mike's - 06/05/2023: Corn, silage

Field name: Mike's Crop: Corn, silage Plant date: 06/05/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	200.38	86.94	116.75	4,769.28
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	352.40
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	247.38	86.94	116.75	5,121.68
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	185.09	28.39	128.31	605.19
Nutrient balance	62.28	58.55	-11.56	4,516.49
Applied to removed ratio	1.34	3.06	0.91	8.46

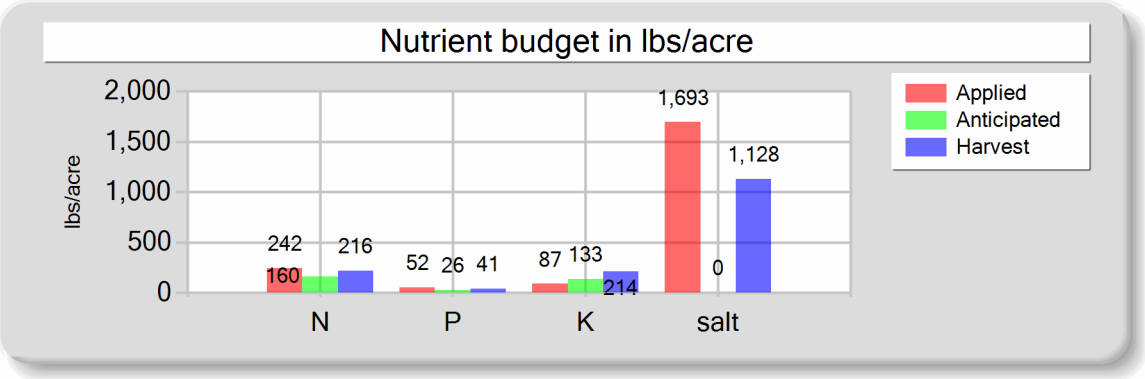
Fresh water applied	
13,545,000.00	*gallons*
498.82	*acre-inches*
29.34	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH1 - 10/13/2022: Oats, silage-soft dough

Field name: RH1 Crop: Oats, silage-soft dough Plant date: 10/13/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	229.72	52.42	87.06	1,566.83
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.78	0.00	0.00	126.47
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	242.50	52.42	87.06	1,693.29
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	215.91	40.60	213.64	1,127.74
Nutrient balance	26.58	11.82	-126.58	565.55
Applied to removed ratio	1.12	1.29	0.41	1.50

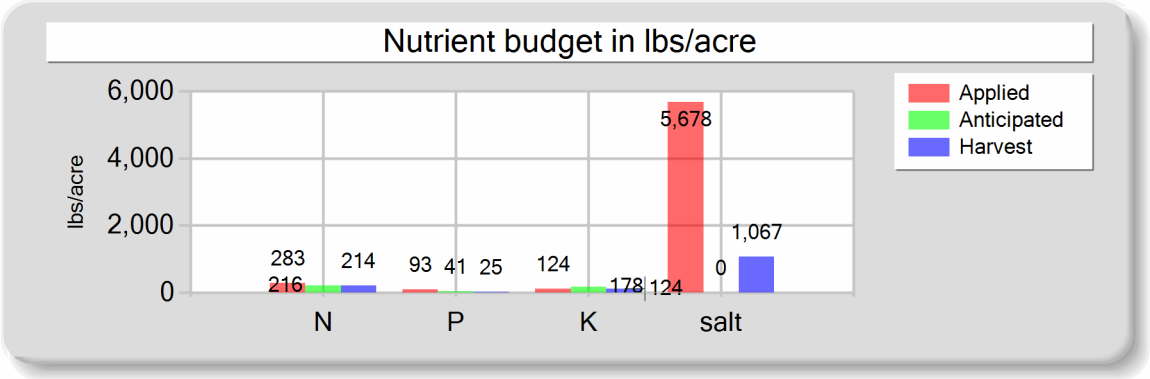
Fresh water applied
7,830,000.00 *gallons*
288.35 *acre-inches*
4.65 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

RH1 - 05/27/2023: Corn, silage

Field name: RH1 Crop: Corn, silage Plant date: 05/27/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	40.00	0.00	0.00	0.00
Dry manure	213.66	92.70	124.49	5,085.52
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.23	0.00	0.00	592.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	282.89	92.70	124.49	5,677.53
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	214.49	25.27	123.77	1,067.39
Nutrient balance	68.41	67.43	0.72	4,610.14
Applied to removed ratio	1.32	3.67	1.01	5.32

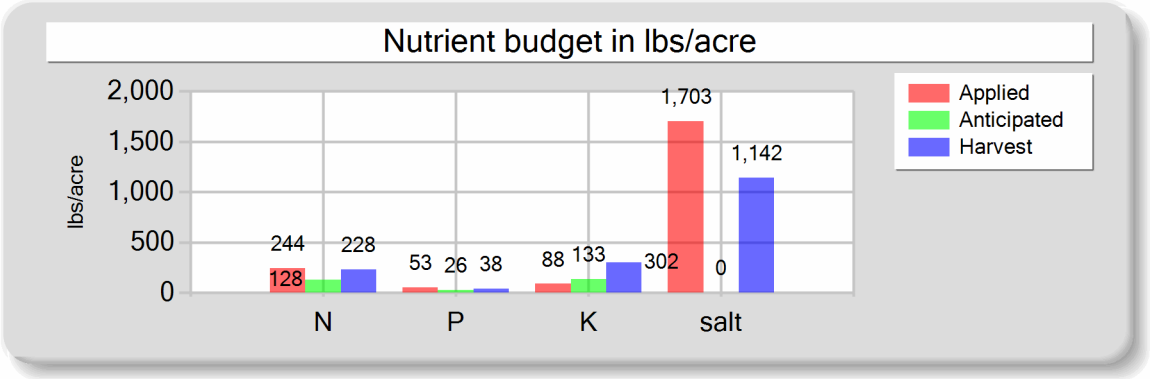
Fresh water applied	
46,791,000.00	*gallons*
1,723.15	*acre-inches*
27.79	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH2 - 10/12/2022: Oats, silage-soft dough

Field name: RH2 Crop: Oats, silage-soft dough Plant date: 10/12/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	231.21	52.76	87.62	1,577.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	5.73	0.00	0.00	125.53
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	243.95	52.76	87.62	1,702.53
Anticipated crop nutrient removal	128.00	25.60	132.80	0.00
Actual crop nutrient removal	228.11	37.58	302.38	1,142.50
Nutrient balance	15.84	15.19	-214.75	560.03
Applied to removed ratio	1.07	1.40	0.29	1.49

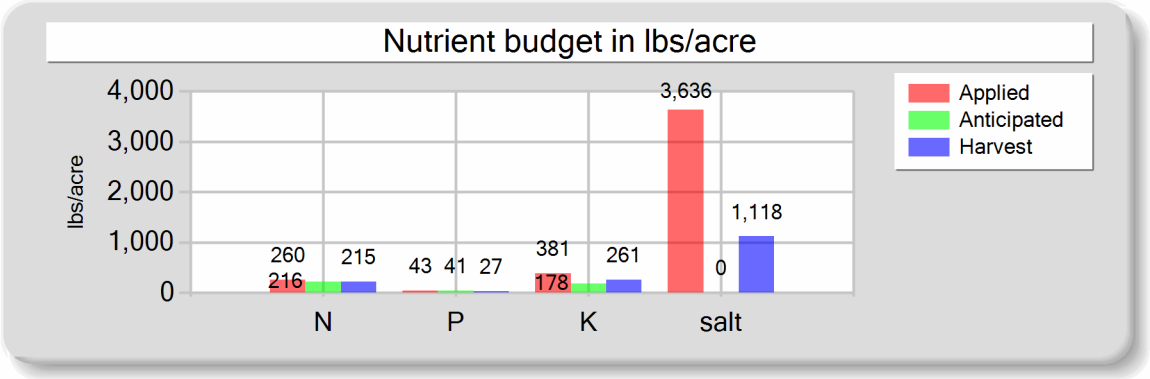
Fresh water applied	
7,020,000.00	*gallons*
258.52	*acre-inches*
4.62	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH2 - 05/23/2023: Corn, silage

Field name: RH2 Crop: Corn, silage Plant date: 05/23/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	222.60	42.93	381.07	3,038.58
Fresh water	22.12	0.00	0.00	597.44
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	259.72	42.93	381.07	3,636.02
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	215.35	26.99	260.51	1,117.88
Nutrient balance	44.37	15.94	120.56	2,518.13
Applied to removed ratio	1.21	1.59	1.46	3.25

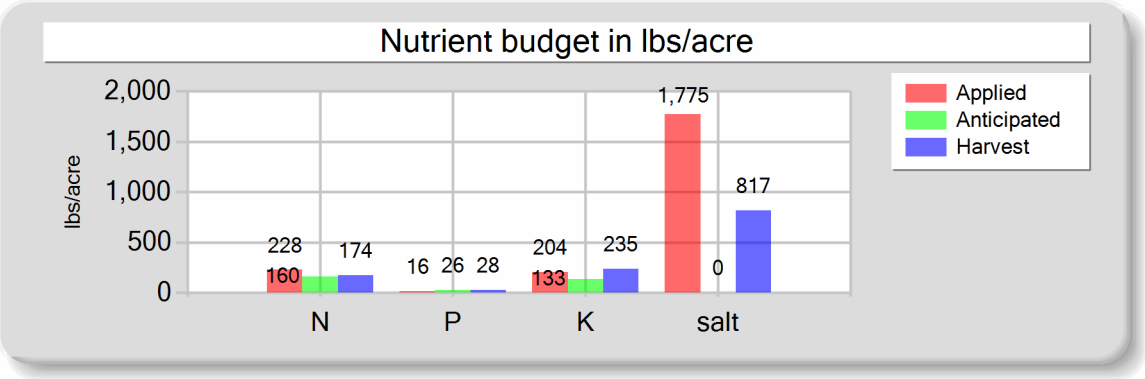
Fresh water applied	
42,444,000.00	*gallons*
1,563.07	*acre-inches*
27.91	*inches/acre*

Process wastewater applied	
2,876,328.00	*gallons*
105.93	*acre-inches*
1.89	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH3 - 10/11/2022: Oats, silage-soft dough

Field name: RH3 Crop: Oats, silage-soft dough Plant date: 10/11/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	215.52	15.65	203.92	1,635.50
Fresh water	5.10	0.00	0.00	139.70
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	227.62	15.65	203.92	1,775.20
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	173.50	28.27	234.51	817.47
Nutrient balance	54.12	-12.62	-30.59	957.73
Applied to removed ratio	1.31	0.55	0.87	2.17

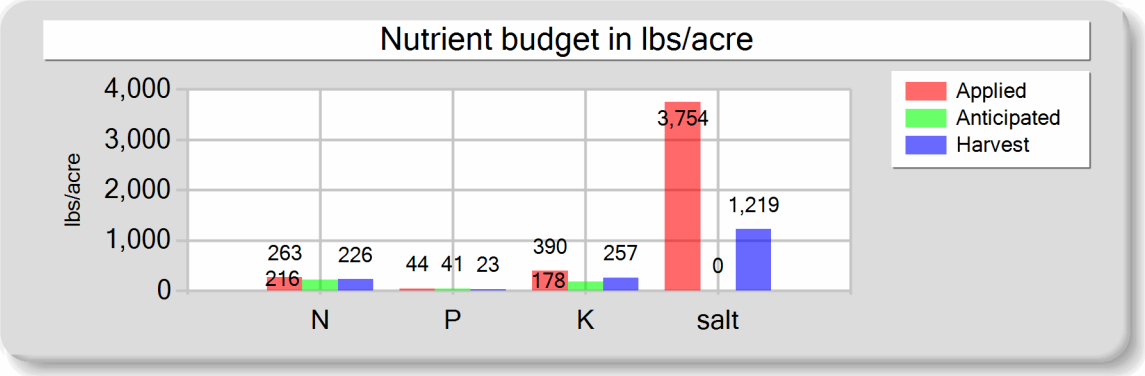
Fresh water applied	
3,780,000.00	*gallons*
139.20	*acre-inches*
3.98	*inches/acre*

Process wastewater applied	
1,765,120.00	*gallons*
65.00	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH3 - 05/22/2023: Corn, silage

Field name: RH3 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.05	43.96	390.06	3,098.74
Fresh water	21.66	0.00	0.00	655.07
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	262.71	43.96	390.06	3,753.80
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	226.25	23.05	257.35	1,219.04
Nutrient balance	36.45	20.91	132.70	2,534.76
Applied to removed ratio	1.16	1.91	1.52	3.08

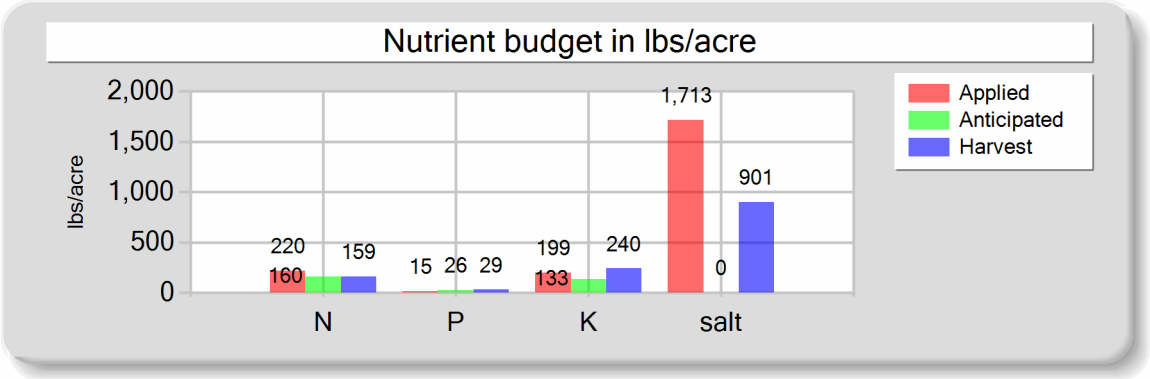
Fresh water applied	
28,035,000.00	*gallons*
1,032.43	*acre-inches*
29.50	*inches/acre*

Process wastewater applied	
1,837,885.00	*gallons*
67.68	*acre-inches*
1.93	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH4 - 10/10/2022: Oats, silage-soft dough

Field name: RH4 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	207.80	15.26	199.09	1,576.20
Fresh water	5.01	0.00	0.00	137.24
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	219.81	15.26	199.09	1,713.44
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	158.92	29.46	239.87	900.94
Nutrient balance	60.89	-14.20	-40.77	812.50
Applied to removed ratio	1.38	0.52	0.83	1.90

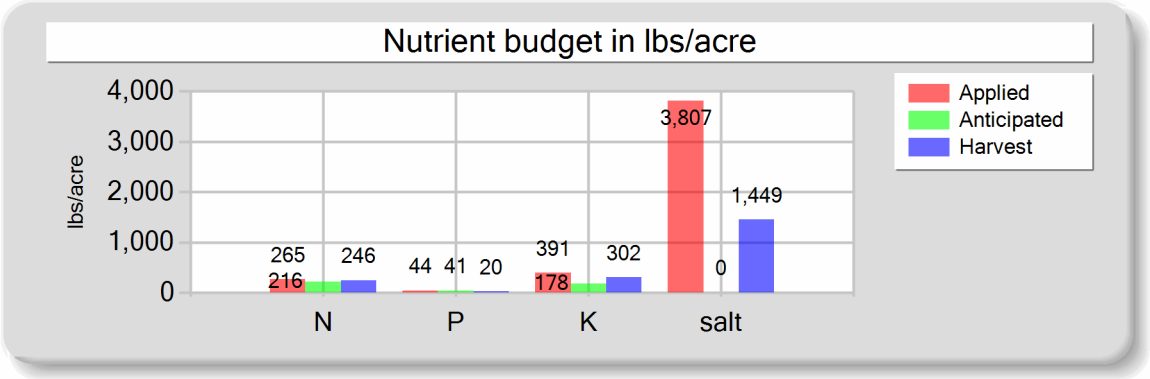
Fresh water applied	
2,016,000.00	*gallons*
74.24	*acre-inches*
3.91	*inches/acre*

Process wastewater applied	
918,327.00	*gallons*
33.82	*acre-inches*
1.78	*inches/acre*

Total harvests for the crop	
1	*harvests*

RH4 - 05/22/2023: Corn, silage

Field name: RH4 Crop: Corn, silage Plant date: 05/22/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.58	44.10	391.25	3,107.20
Fresh water	23.13	0.00	0.00	699.61
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	264.72	44.10	391.25	3,806.82
Anticipated crop nutrient removal	216.00	40.50	178.20	0.00
Actual crop nutrient removal	246.08	20.38	301.84	1,449.43
Nutrient balance	18.63	23.72	89.41	2,357.38
Applied to removed ratio	1.08	2.16	1.30	2.63

Fresh water applied	
16,254,000.00	*gallons*
598.58	*acre-inches*
31.50	*inches/acre*

Process wastewater applied	
1,000,654.00	*gallons*
36.85	*acre-inches*
1.94	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Separator Solids

Sample and source description: Separator Solids

Sample date: 09/13/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 66.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	4,450.00	7,390.00							13.30
DL	100.00	100.00	100.00							0.01

Compost

Sample and source description: Compost

Sample date: 04/18/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 17.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,300.00	7,300.00	6,400.00							45.30
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/18/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 60.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	24,200.00	10,500.00	14,100.00							57.60
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 04/18/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,900.00	8,400.00							7.01
DL	100.00	100.00	100.00							0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,900.00	6,510.00	17,300.00							25.40
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 70.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,900.00	2,650.00	5,800.00							7.98
DL	100.00	100.00	100.00							0.01

Separator Solids

Sample and source description: Separator Solids

Sample date: 10/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 57.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,400.00	3,680.00	7,420.00							16.10
DL	100.00	100.00	100.00							0.01

B. PROCESS WASTEWATER ANALYSES

4QWW

Sample and source description: 4QWW

Sample date: 10/25/2022 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	645.00	494.00	0.00	0.00	64.90	875.00								8,420.00	4,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

1QWW

Sample and source description: 1QWW

Sample date: 02/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	386.00	224.00	0.00	0.00	10.90	114.00	38.50	22.20	40.60	0.00	0.00	160.00	260.00	4,760.00	3,000
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

2QWW

Sample and source description: 2QWW

Sample date: 04/11/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	753.00	448.00	0.00	0.00	89.30	817.00								7,710.00	8,280
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 07/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	868.00	675.00	0.00	0.00	61.10	883.00								10.10	6,060
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

3QWW

Sample and source description: 3QWW

Sample date: 10/04/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	286.00	143.00	0.00	0.00	111.00	961.00								9,390.00	5,900
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/25/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	645.00	494.00	0.00	0.00	64.90	875.00								8,420.00	4,820
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

4QWW

Sample and source description: 4QWW

Sample date: 10/31/2023 Material type: Process wastewater Source of analysis: Lab analysis pH:

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	809.00	406.00	0.00	0.00	94.30	758.00								6,490.00	4,760
DL	0.50	0.50	0.50	0.50	0.50	0.50								1.00	10

C. FRESH WATER ANALYSES

Canal Woodbridge

Fresh Water

Sample description: Fresh Water

Sample date: 08/17/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	0.00	0.00	0.00								66.60	53
DL	0.50	0.50	0.50								10.00	10

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.46	0.00	5.46								200.00	122
DL	0.50	0.50	0.50								1.00	10

IW 2

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.44	0.00	9.44								256.00	130
DL	0.50	0.50	0.50								1.00	10

IW 3

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	8.47	0.00	8.47								288.00	155
DL	0.50	0.50	0.50								1.00	10

IW 4

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	14.70	0.00	14.70								468.00	267
DL	0.50	0.50	0.50								1.00	10

IW 5

IW 5

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.92	0.00	5.92								602.00	313
DL	0.50	0.50	0.50								1.00	10

IW 6

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.24	0.00	3.24								161.00	98
DL	0.50	0.50	0.50								1.00	10

IW 7

Fresh Water

Sample description: Fresh Water

Sample date: 09/08/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.53	0.00	3.53								155.00	94
DL	0.50	0.50	0.50								1.00	10

IW Fontes

IW Fontes

Fresh Water

Sample description: Fresh Water

Sample date: 09/21/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.63	0.00	7.63								1,220.00	690
DL	0.50	0.50	0.50								1.00	10

D. SOIL ANALYSES

D1

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			32.00				
DL			4.00				

D2

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			68.00				
DL			4.00				

D3

D3

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			108.00				
DL			4.00				

D4

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			90.00				
DL			4.00				

D5

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			79.00				
DL			4.00				

D6

D6

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			70.00				
DL			4.00				

Fontes

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			43.00				
DL			4.00				

H1

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			46.00				
DL			4.00				

H2

H2

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			48.00				
DL			4.00				

H3

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			55.00				
DL			4.00				

Jennifer's

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00				
DL			4.00				

Mike's

Mike's

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00				
DL			4.00				

RH1

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			115.00				
DL			4.00				

RH2

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			110.00				
DL			4.00				

RH3

RH3

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

RH4

Soils

Sample and source description: Soils

Sample date: 11/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

D1 - 11/04/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,770.00	900.00	5,130.00		7.26
DL	100.00	100.00	100.00		0.01

D1 - 06/05/2023: Corn, silage

D1 - 06/05/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,400.00	640.00	3,300.00		4.40
DL	100.00	100.00	100.00		0.01

D2 - 10/17/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,620.00	1,620.00	21,100.00		13.10
DL	100.00	100.00	100.00		0.01

D2 - 06/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 79.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,100.00	270.00	2,720.00		5.10
DL	100.00	100.00	100.00		0.01

D3 - 10/18/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 04/18/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	8,310.00	2,040.00	27,200.00		12.60
DL	100.00	100.00	100.00		0.01

D3 - 06/21/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/17/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 78.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	2,530.00	270.00	2,250.00		3.90
DL	100.00	100.00	100.00		0.01

D4 - 10/17/2022: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,700.00	3,110.00	19,200.00		9.40
DL	100.00	100.00	100.00		0.01

D5 - 10/15/2019: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 8.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,900.00	2,810.00	43,300.00		13.10
DL	100.00	100.00	100.00		0.01

D6 - 09/17/2020: Alfalfa, hay

Alfalfa

Sample and source description: Alfalfa

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 7.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,600.00	2,820.00	27,100.00		10.70
DL	100.00	100.00	100.00		0.01

Fontes - 01/10/2018: Almond Field Run

Almonds

Sample and source description: Almonds

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 18.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	15,800.00	1,230.00	16,700.00		10.50
DL	100.00	100.00	100.00		0.01

H1 - 10/22/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 59.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,050.00	910.00	5,160.00		6.45
DL	100.00	100.00	100.00		0.01

H1 - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 64.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,790.00	640.00	3,930.00		4.00
DL	100.00	100.00	100.00		0.01

H2 - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 56.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,220.00	1,160.00	6,490.00		7.16
DL	100.00	100.00	100.00		0.01

H2 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 65.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,080.00	750.00	3,350.00		3.80
DL	100.00	100.00	100.00		0.01

H3 - 10/20/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,200.00	1,100.00	5,720.00		6.55
DL	100.00	100.00	100.00		0.01

H3 - 06/07/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,480.00	610.00	2,940.00		4.60
DL	100.00	100.00	100.00		0.01

Jennifer's - 10/21/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 63.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,960.00	1,020.00	5,480.00		7.08
DL	100.00	100.00	100.00		0.01

Jennifer's - 06/03/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,390.00	520.00	2,350.00		3.40
DL	100.00	100.00	100.00		0.01

RH1 - 10/13/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,690.00	1,070.00	5,630.00		7.78
DL	100.00	100.00	100.00		0.01

RH1 - 05/27/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,310.00	390.00	1,910.00		5.68
DL	100.00	100.00	100.00		0.01

RH2 - 10/12/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 54.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	7,770.00	1,280.00	10,300.00		8.61
DL	100.00	100.00	100.00		0.01

RH2 - 05/23/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 69.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,910.00	490.00	4,730.00		6.59
DL	100.00	100.00	100.00		0.01

RH3 - 10/11/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 70.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,830.00	950.00	7,880.00		9.28
DL	100.00	100.00	100.00		0.01

RH3 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 67.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,220.00	430.00	4,800.00		6.89
DL	100.00	100.00	100.00		0.01

RH4 - 10/10/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/03/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 68.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	5,340.00	990.00	8,060.00		9.52
DL	100.00	100.00	100.00		0.01

RH4 - 05/22/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 09/19/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 50.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	6,400.00	530.00	7,850.00		7.60
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

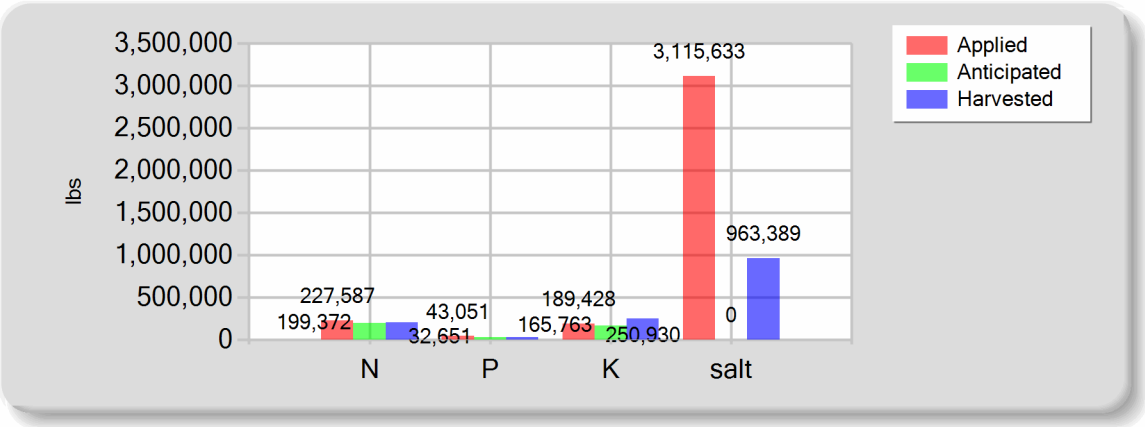
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

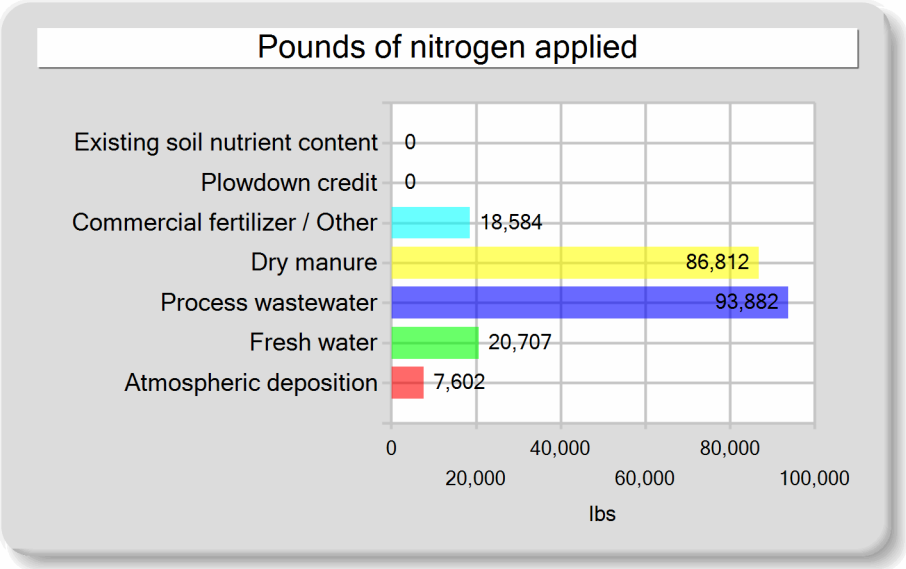
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

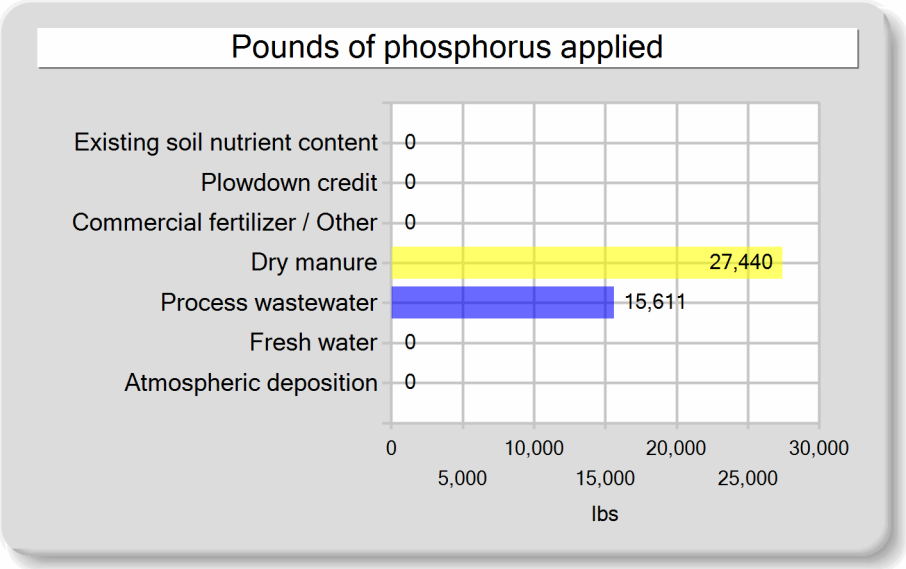
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	18,584.00	0.00	0.00	0.00
Dry manure	86,812.14	27,439.99	40,454.08	1,221,964.80
Process wastewater	93,882.18	15,610.98	148,974.34	1,091,555.43
Fresh water	20,706.58	0.00	0.00	802,113.16
Atmospheric deposition	7,602.00	0.00	0.00	0.00
Total nutrients applied	227,586.90	43,050.98	189,428.41	3,115,633.40
Anticipated crop nutrient removal	199,372.00	32,650.70	165,763.00	0.00
Actual crop nutrient removal	205,541.20	31,373.05	250,930.37	963,388.85
Nutrient balance	22,045.70	11,677.93	-61,501.96	2,152,244.55
Applied to removed ratio	1.11	1.37	0.75	3.23

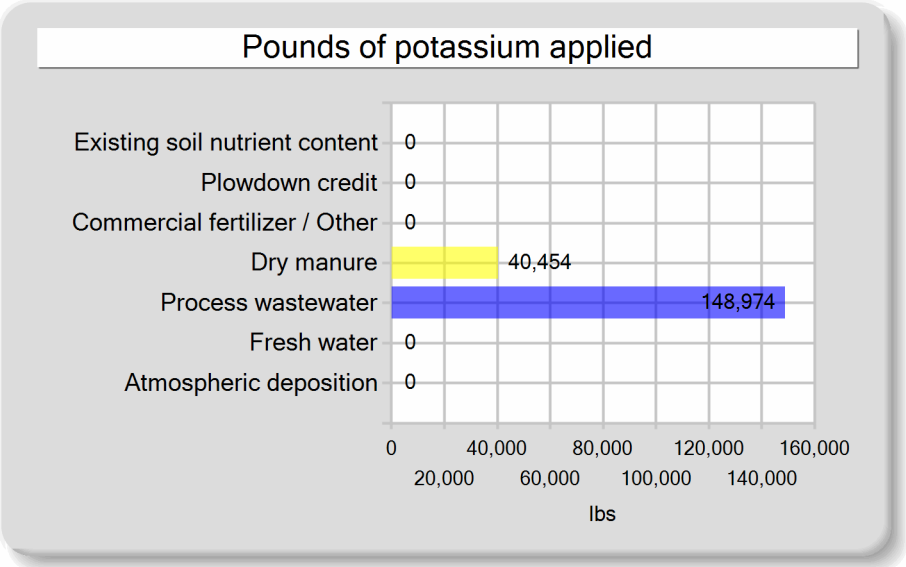
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

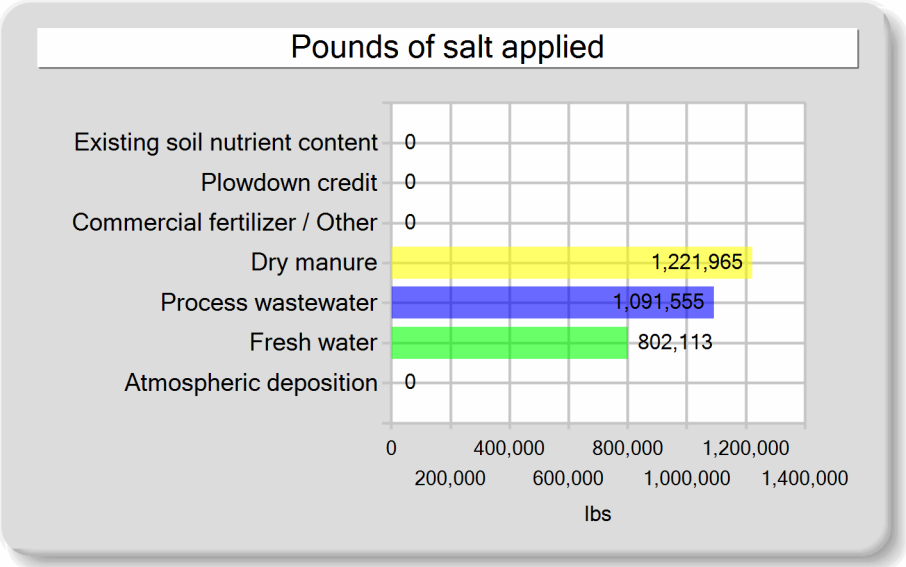


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Field D2 Corn (1.52) had a high nitrogen field ratio as crop failed to meet anticipated nitrogen removal of 216 lbs./acre. The corn crop removed only 135.7 pounds of nitrogen/acre as yield was adequate however a low nitrogen value from the plant tissue lab analysis resulted in only 4.2 pounds of nitrogen removed per harvested ton compared to the anticipated removal of 8 pounds of nitrogen per harvested ton. Nitrogen application (201.6 pounds) was not considered excessive but in line with that recommended in the dairy's nutrient management plan.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Jack Hamm
Jack Hamm (Mar 5, 2024 10:18 PST)

SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Jack A Hamm	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Mar 5, 2024	
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Jack Hamm

Name of Dairy Facility: Lima Ranch

Facility Address: 13436 N. Thornton Road (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person Name and Phone Number: Jack Hamm (Name) 209-969-5225 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Albert Cunha

Address of Hauling Company /Person: 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person: Albert Cunha (Name) 209-334-1851 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Albert Cunha (Name) 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) 209-334-1851 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 2/18/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2100 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 17.9% moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

210 loads at 10 ton/load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Jack Hamm (Jack Hamm (Mar 5, 2024 10:18 PST)) Date: Mar 5, 2024

Hauler's Signature: Robert Cinha Date: 2-15-24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Jack Hamm

Name of Dairy Facility: Lima Ranch

Facility Address: 13436 N. Thornton Road (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person Name and Phone Number: Jack Hamm (Name) 209-969-5225 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Albert Cunha

Address of Hauling Company /Person: 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person: Albert Cunha (Name) 209-334-1851 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Albert Cunha (Name) 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) 209-334-1851 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 8/21/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 3000 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 57.1% moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

300 loads at 10 ton/load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Jack Hamm (Jack Hamm (Mar 5, 2024 10:18 PST)) Date: Mar 5, 2024

Hauler's Signature: Albert Cinha Date: 2-15-24

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination**.
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: Jack Hamm

Name of Dairy Facility: Lima Ranch

Facility Address: 13436 N. Thornton Road (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person Name and Phone Number: Jack Hamm (Name) 209-969-5225 (Phone Number)

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Albert Cunha

Address of Hauling Company /Person: 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code)

Contact Person: Albert Cunha (Name) 209-334-1851 (Phone Number)

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) _____ (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Albert Cunha (Name) 1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) 209-334-1851 (Phone Number)

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

1327 W. Harney Lane, (Number and Street) Lodi, CA (City) 95242 (Zip Code) _____ (Assessor's Parcel Number)

Dates Hauled: 10/24/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 2000 Tons or Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 57.1% moisture

Manure Density (if amount reported in cubic yards): _____

Method used to determine amount of manure: _____

200 loads at 10 ton/load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification E.3 of Reissued Waste Discharge Requirements General Order No. R5-2013-0122) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: Jack Hamm (Jack Hamm (Mar 5, 2024 10:18 PST)) Date: Mar 5, 2024

Hauler's Signature: Robert Cinha Date: 2-15-24

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

nase Order No

Bill To: Acct # 0 Cons # 13

No. Samples: No of Bottles:

Results Need By

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Name: LIMA Ranch

Address: 13436 N Thornton Rd

City: Lodi State: CA Zip: 95242

Telephone: 209 334-5422 Fax:

Cell/Email:

COPY TO: F&A Ag Services
& JSawyer@Fragservices.com

REQUESTED BY: Jeanette Sawyer

PROJECT:

CROP:

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DWW1

[ ] Co. Health Dept

[ ] RWQCB [ ] Copy of Chain

[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	DW 1	9/8/23			
2	2				
3	3				
4	4				
5	5				
6	6				
7	7				
8	8				
9	9				
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

mpling hrs

es

Consulting

Amt Paid	Rec By	Check #	Date

Shipping

$ In

$ Out

Signature

Sample received in cooler with ice (coolant)

[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

Report of Dairy Well Water Analysis

Lima Ranch
13436 N Thorton Rd
Lodi CA 95242
00-0026186 13

E-mail: jswayer@fragservices.com
Copy To:

Lab No.: 23I0653
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 09/08/23
Reported Date: 09/18/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	TKN mg/L	Total Nitrogen TN mg/L	pH at 25°C unit
1	DW 1	9/7/23	7:30	411	0.41	1.5	*		ND	1.50	7.5
2	DW 2	9/7/23	7:30	482	0.48	4.4	*		ND	4.44	7.4
3	DW 3	9/7/23	7:30	322	0.32	1.8	*		ND	1.82	7.6
4	DW 4	9/7/23	7:30	434	0.43	4.2	*		ND	4.30	7.5
5	DW 5	9/7/23	7:30	339	0.34	1.1	*		ND	1.21	7.6
6	DW 6	9/7/23	7:30	325	0.32	4.5	*		ND	4.54	7.6
7	DW 7	9/7/23	7:30	454	0.45	13.7	*		ND	13.8	7.5
8	DW 8	9/7/23	7:30	118	0.12	2.0	*		ND	1.98	7.6
9	DW 9	9/7/23	7:30	132	0.13	3.0	*		ND	3.00	7.4

* = Field NH4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Purchase Order No

Bill To: Acct # 0 Cons # 13

Results Need By

No. Samples: No of Bottles:

Name: LiMA Ranch
Address: 13436 N Thornton Rd
City: Lodi **State:** CA **Zip:** 95242
Telephone: 209 334-5422 **Fax:**
Cell/Email:
COPY TO: F&R Ag Services
& JSawyer@Frogservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCWI

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW 2	9/8/23			
2	3				
3	4				
4	5				
5	6				
6	7				
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (*cal*). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through *cal* under its Rules and Procedures. The parties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling hrs
Miles
Consulting

Shipping
$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE™
LABORATORY INC

Report of Dairy Canal Water Analysis

Lima Ranch
13436 N Thorton Rd
Lodi CA 95242
00-0026186 13

E-mail: jswayer@fragservices.com
Copy To:

Lab No.: 23I0651
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 09/08/23
Reported Date: 09/14/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	TKN mg/L	Total Nitrogen TN mg/L
1	IW 2	9/7/23	6:30	201	0.20	5.4	122	ND	5.46
2	IW 3	9/7/23	6:30	288	0.29	8.4	155	ND	8.47
3	IW 4	9/7/23	6:30	468	0.47	14.6	267.0	ND	14.7
4	IW 5	9/7/23	6:30	602	0.60	5.9	313	ND	5.92
5	IW 6	9/7/23	6:30	161	0.16	3.1	98	ND	3.24
6	IW 7	9/7/23	6:30	155	0.15	2.9	94	ND	3.53

ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

WATER WORK REQUEST

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728
www.dellavallelab.com 559 233-6129 · 800 228-9896 · Fax 559 268-8174

Chase Order No

Bill To: Acct # 0 Cons # 13

Results Need By

No. Samples: No of Bottles:

Water Type: [ ] Drinking Water [ ] Wastewater
[ ] Ag Water [ ] Groundwater [ ] Monitoring Well

Other:

Name: Lima Ranch
Address: 13436 N Thornton Rd
City: Lodi **State:** CA **Zip:** 95242
Telephone: 209 334-5422 **Fax:**
Cell/Email:
COPY TO: F&R Ag Services
& JSawyer@Frogservices.com
REQUESTED BY: Jeanette Sawyer
PROJECT:
CROP:

Analysis and Bottles Required: (*Please indicate Analysis*)

DCW1

[ ] Co. Health Dept
[ ] RWQCB [ ] Copy of Chain
[ ] State Forms [ ] QA/QC Documents

Sampled By:

	Description of Samples	Date Sampled	Time Sampled	Rec'd Temp °C	Field EC
1	IW Fontes	9/21/23	7AM		
2					
3					
4					
5					
6					
7					
8					
9					
10					

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	Jeanette Sawyer	F&R Ag Services		
Second				
Third				
Fourth				

I guarantee that as the client, or on behalf of client named, I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a liquidated damage fee of 2% per month (annually 24%) or $5.00 per month whichever is greater.

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Invoicing Information:

Sampling hrs
Fees
Consulting

Shipping

$ In
$ Out

Amt Paid	Rec By	Check #	Date

Signature

Sample received in cooler with ice (coolant)
[ ] Yes [ ] No

DELLAVALLE LABORATORY INC

Report of Dairy Can. Water Analysis

Lima Ranch
13436 N Thorton Rd
Lodi CA 95242
00-0026186 13

E-mail: jswayer@fragservices.com
Copy To:

Lab No.: 23I1419
Sampled By:
Requested By: Jeanette Sawyer
Submitted Date: 09/21/23
Reported Date: 01/26/24
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	TDS mg/L	Total TKN mg/L	Total Nitrogen TN mg/L
1	IW Fontes	09/21/23	7:00	1220	1.22	7.6	690	ND	7.63

ND = None Detected

Approved By: Scott McFriedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

Lima Ranch 2023 Annual Report

Final Audit Report 2024-03-05

Created:	2024-03-04
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAATM9P-pcqnjProMQA9buB2fa-Ax8y8vzQ

"Lima Ranch 2023 Annual Report" History

Document created by Vince Furtado (vfurtado@fragservices.com)
2024-03-04 - 4:40:24 PM GMT

Document emailed to Jennifer Hamm (mooovers@aol.com) for signature
2024-03-04 - 4:43:03 PM GMT

Email viewed by Jennifer Hamm (mooovers@aol.com)
2024-03-04 - 5:02:03 PM GMT

Signer Jennifer Hamm (mooovers@aol.com) entered name at signing as Jack Hamm
2024-03-05 - 6:18:20 PM GMT

Document e-signed by Jack Hamm (mooovers@aol.com)
Signature Date: 2024-03-05 - 6:18:22 PM GMT - Time Source: server

Agreement completed.
2024-03-05 - 6:18:22 PM GMT

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